Can Time Series Foundation Models Forecast Realized Volatility?

Alessio Brini
Duke University
alessio.brini@duke.edu

Abstract

Realized volatility forecasting is central to risk management and derivative pricing, yet no study has systematically assessed whether pretrained time series foundation models (TSFMs) can match established econometric benchmarks for this task. We evaluate nine zero-shot TSFMs against six econometric specifications on 50 assets from the VOLARE dataset at multiple forecast horizons, using formal statistical tests for pairwise and multi-model forecast comparison. The best TSFMs outperform the best econometric benchmarks across all horizons and asset classes. Sundial and TTM achieve the lowest quasi-likelihood (QLIKE) loss, with Sundial leading at daily and weekly horizons and TTM at the monthly horizon for equities (QLIKE = 0.190 vs. Sundial's 0.218). Sundial enters the Model Confidence Set for the vast majority of assets across all horizons; its advantage over the best econometric specification (Log-HAR) grows from 11% at $h = 1$ to 37% at $h = 22$. TTM, the smallest model in the evaluation (<1M parameters), enters the MCS for most assets at the monthly horizon despite failing to enter at the daily horizon. However, performance is highly heterogeneous across architectures: some models produce forecasts with severe scale misalignment relative to realized values, and model selection within the TSFM class matters as much as the choice between TSFMs and econometric benchmarks. Subject to the caveat that indirect data contamination cannot be fully excluded, these results suggest that zero-shot TSFMs are a viable alternative to HAR-family models, provided practitioners evaluate multiple architectures rather than adopting a single TSFM by default.

Keywords: Realized volatility, heterogeneous autoregressive model, time series foundation models, zero-shot forecasting
JEL Codes: C22, C45, C53, G17

1 Introduction

Forecasting realized volatility is central to risk management, derivative pricing, and portfolio allocation. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has become the standard model-free measure of ex-post price variation. The Heterogeneous Autoregressive (HAR) model of Corsi (2009) is the dominant forecasting benchmark: its three-component structure, aggregating past realized volatility at daily, weekly, and monthly frequencies, provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Extensions incorporating jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016) have yielded mixed practical gains (Clements and Preve, 2021; Branco et al., 2024). Machine learning methods, including neural networks (Bucci, 2020), random forests (Luong and Dokuchaev, 2018), graph-based approaches (Zhang et al., 2024, 2025; Brini and Toscano, 2025), and convolutional architectures (Moreno-Pino and Zohren, 2024), have also been applied, but a broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements rather than a paradigm shift.

A different class of models has emerged in recent years: time series foundation models (TSFMs). These are large pretrained transformer neural networks (Vaswani et al., 2017), architectures that learn dependencies between positions in a sequence through an attention mechanism without imposing a fixed lag structure, trained on large and diverse corpora of time series from multiple domains, capable of producing forecasts for previously unseen series without any retraining, a capability known as zero-shot forecasting. Leading examples include Chronos (Ansari et al., 2024), Moirai (Woo et al., 2024), and Lag-Llama (Rasul et al., 2024), several of which have released second-generation versions with improved architectures (Ansari et al., 2025; Liu et al., 2025a). Surveys document the rapid pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024). Outside finance, Carriero et al. (2024) applied zero-shot large language models and time series foundation models to

macroeconomic forecasting and found that they do not outperform traditional econometric baselines, raising the question of whether similar conclusions hold for financial time series.

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM 1.0 on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. However, that negative result may be model-vintage-specific rather than a fundamental limitation of zero-shot inference: TimesFM 2.5, released one year later, performs well zero-shot in our evaluation. Rahimikia et al. (2025) evaluated several TSFMs on daily stock returns and reported uniformly negative zero-shot results. Realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits well-documented mean reversion and long-memory dependence (Andersen et al., 2001, 2003), and follows a distribution closer to the macroeconomic and physical series on which TSFMs were pretrained.

To our knowledge, no study has yet evaluated multiple foundation models on realized volatility with formal statistical testing. This paper conducts the first systematic comparison of zero-shot TSFMs against established econometric benchmarks for realized volatility forecasting. We evaluate nine TSFMs, spanning eight distinct architectures, against six econometric specifications across 50 assets spanning three asset classes (equities, FX, futures) and three forecast horizons ($h = 1, 5, 22$ days), using the VOLARE dataset (Cipollini et al., 2026). The zero-shot setting, with no domain-specific training, isolates the value of general time series pretraining. We apply Diebold–Mariano (DM) tests (Diebold and Mariano, 1995) and Model Confidence Set (MCS) analysis (Hansen et al., 2011) to control for multiple comparisons, and supplement these with Mincer–Zarnowitz regressions (Mincer and Zarnowitz, 1969) for forecast efficiency, Giacomini–Rossi fluctuation tests (Giacomini and Rossi, 2010) for time-varying relative performance, sub-sample analysis across pre- and post-COVID regimes, and context window sensitivity checks.

The best zero-shot TSFMs outperform the best econometric benchmarks across all horizons and asset classes, contrasting with prior negative findings on daily returns and on

TimesFM 1.0. Sundial, a flow-matching generative model,[1] achieves the lowest QLIKE at short horizons and enters the MCS for 92–100% of assets, while TTM, the smallest model in the evaluation (<1M parameters), achieves the lowest equity QLIKE at the monthly horizon. The TSFM advantage grows with forecast horizon, reaching its largest margin at the monthly horizon. However, performance is highly heterogeneous across TSFM architectures: some models produce forecasts with severe scale misalignment while others outperform all benchmarks, indicating that model selection within the TSFM class is as important as the choice between TSFMs and econometric models.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the VOLARE dataset. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results. Section 6 assesses statistical significance through DM tests, MCS, Mincer–Zarnowitz regressions, and Giacomini–Rossi fluctuation tests. Section 7 contains robustness checks, and Section 8 concludes.

2 Literature Review

2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean

[1] Flow matching is a generative modeling technique that learns continuous transformations from a noise distribution to the target forecast distribution; see Section 4.2 for details.

reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding to heterogeneous trading horizons. The specification is parsimonious (three regressors plus a constant) yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. The HAR-J model (Andersen et al., 2007) separates realized variance into continuous and jump components using bipower variation (Barndorff-Nielsen and Shephard, 2004). The HAR-RS model (Patton and Sheppard, 2015; Barndorff-Nielsen et al., 2010) decomposes realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. The HARQ model (Bollerslev et al., 2016) interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. Autoregressive Fractionally Integrated Moving Average (ARFIMA) models (Granger and Joyeux, 1980; Hosking, 1981), applied to realized volatility by Andersen et al. (2003), explicitly parameterize the fractional integration order and remain competitive at longer forecast horizons where the slow mean reversion of realized volatility becomes the dominant dynamic.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network

that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). TSFMs differ from task-specific deep learning models (neural networks, recurrent networks, convolutional architectures) in a fundamental way: deep learning models must be trained from scratch on the target series before they can produce any forecast, while TSFMs are pretrained on large external corpora and can forecast immediately via zero-shot inference. This eliminates the need for domain-specific training data, hyperparameter tuning, and GPU-intensive estimation, a practical advantage when deploying to new assets or markets where historical data may be limited. TSFMs are trained on vast corpora of time series drawn from diverse domains (weather, energy, retail, transport, macroeconomics) and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. T5, the Text-to-Text Transfer Transformer (Raffel et al., 2020), is a sequence-to-sequence model that reads the full input through an encoder and generates output tokens autoregressively through a decoder. Moirai (Woo et al., 2024) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification. Lag-Llama (Rasul et al., 2024) adapts the LLaMA decoder-only architecture for probabilistic time series forecasting using lag-based tokenization. Finance-specific foundation models have also been

proposed, including Kronos (Shi et al., 2025), which is pretrained on candlestick (open-high-low-close-volume, OHLCV) data from over 45 global exchanges using a learned tokenizer that maps price patterns into discrete tokens.

Second-generation models appeared in 2025. Ansari et al. (2025) introduced Chronos-2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while Chronos-Bolt is a distilled variant, meaning its parameters were trained to approximate the full Chronos-2 model's behavior at a fraction of the computational cost, producing direct quantile forecasts in a single forward pass and running up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors using multi-token prediction and improved tokenization. Other models include TimesFM (Das et al., 2024), a patch-based decoder model from Google; Toto (Cohen et al., 2024), a model from Datadog pretrained on infrastructure monitoring metrics (e.g., server CPU usage, request latency); Moirai-MoE (Liu et al., 2024a), a sparse Mixture of Experts extension of Moirai; Sundial (Liu et al., 2025b), which uses flow matching for generative forecasting; MOMENT (Goswami et al., 2024), a masked-encoder architecture designed for multiple time series tasks (forecasting, classification, anomaly detection); and Timer (Liu et al., 2024b).

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate nine foundation models spanning eight distinct architectures (Chronos-Bolt-Small, Chronos-Bolt-Base, Moirai 2.0, Moirai-MoE, Lag-Llama, TimesFM 2.5, Toto, Sundial, and TTM), use newer model generations, and assess forecast significance and robustness through formal statistical testing. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT[2] on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approximately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics, properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and Kronos (Shi et al., 2025).[3]

[2]We do not evaluate MOMENT because it is pretrained with a masked reconstruction objective rather than an autoregressive forecasting objective. Its forecasting head, a linear projection layer that maps patch embeddings to the forecast horizon, is not pretrained and must be trained on the target series before the model can produce any predictions (Goswami et al., 2024, Sections 3.3–3.4). This makes it incompatible with our zero-shot evaluation protocol.

[3]We do not evaluate Kronos because it expects OHLCV price data as input and cannot be meaningfully applied to univariate realized variance without constructing synthetic candlestick data, which introduces an arbitrary input transformation unrelated to the model's intended use case.

The gap in this literature, the absence of a multi-model, multi-asset evaluation with formal statistical testing for realized volatility, motivates the empirical design we describe next.

3 The VOLARE Dataset

We use the VOLARE (VOLatility Archive for Realized Estimates) dataset of Cipollini et al. (2026), which provides daily realized volatility measures for a broad cross-section of financial assets. VOLARE is constructed from ultra-high-frequency tick data sourced from Kibot, covering 40 U.S. equities, 5 major currency pairs, and 5 commodity and index futures contracts. The equity sample begins on January 2, 2015 and runs through January 30, 2026, yielding 2,786 trading days per stock. The FX and futures samples begin on September 25, 2009 (4,242 and 4,224 trading days, respectively).

For each asset-day, VOLARE provides realized measures computed at multiple sampling frequencies. All measures used in this paper are computed at the 5-minute sampling frequency: realized variance (RV), bipower variation (BPV), positive and negative realized semivariance (RS^+, RS^-), and realized quarticity (RQ). All values are expressed in decimal squared returns; for instance, a typical daily RV for a U.S. equity is approximately 2.5×10^{-4}, corresponding to an annualized volatility of roughly 25%. VOLARE is well suited for our multi-asset comparison because it constructs all realized measures using a uniform methodology across assets, ensuring that cross-asset differences in model rankings reflect genuine forecasting performance rather than measurement inconsistencies. The sample period (2015–2026 for equities, 2009–2026 for FX and futures) spans multiple volatility regimes, including the low-volatility period of 2017–2019, the COVID-19 shock of 2020, and the subsequent recovery, providing a sufficiently rich evaluation environment. The equity sample comprises the 40 stocks in VOLARE that span the 2015–2026 period: AAPL, ADBE, AMD, AMGN, AMZN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GE, GOOGL, GS, HD, HON, IBM, JNJ,

JPM, KO, MCD, META, MMM, MRK, MSFT, NFLX, NKE, NVDA, ORCL, PG, PM, SHW, TRV, TSLA, UNH, V, VZ, WMT, and XOM. These cover all 11 Global Industry Classification Standard (GICS) sectors. The FX sample consists of five major currency pairs (AUDUSD, EURUSD, GBPUSD, USDCAD, USDJPY) and the futures sample covers five contracts (Corn, Crude Oil, E-mini S&P 500, Gold, Natural Gas). Both FX and futures samples span 2009–2026, providing substantially longer histories than the equity sample because VOLARE's intraday coverage for these asset classes begins earlier.

Table 1 reports descriptive statistics for the 5-minute realized variance across the three asset classes. As expected for a strictly positive quantity, realized variance is right-skewed across all asset classes, with heavy tails (kurtosis ranging from 35 to over 3,000) confirming that extreme volatility episodes are a persistent feature of the data. The first-order autocorrelation of daily RV, denoted ρ_1, averages 0.598 for equities, consistent with the well-documented persistence of volatility; the 22-day autocorrelation (ρ_{22}, i.e., the sample autocorrelation at lag 22) averages 0.098, indicating slow decay. The FX pairs display RV approximately two orders of magnitude smaller than equities (cross-sectional average $\bar{RV} \approx 3 \times 10^{-5}$ vs. 2.5×10^{-4}), with comparable persistence (cross-sectional average $\bar{\rho}_1 = 0.52$, where the bar denotes averaging across assets). Futures exhibit the widest cross-asset heterogeneity: the E-mini S&P 500 (ES) has $\rho_1 = 0.779$, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$), presenting a natural stress test for forecasting models.

Table 1: Descriptive statistics for daily realized variance measured at the 5-minute frequency. The equity panel reports cross-sectional averages across 40 stocks; FX and futures report individual assets. ρ_k denotes the sample autocorrelation at lag k (e.g., ρ_{22} is the autocorrelation at 22 trading days). Mean and Median are in decimal squared returns ($\times 10^{-4}$); Skewness, Kurtosis, and autocorrelations are dimensionless.

	N	Mean	Median	Skew	Kurt	ρ_1	ρ_{22}
Panel A: Equities (cross-sectional average, 40 stocks, 2015–2026)							
Average	2,786	2.51	1.51	12.6	245	0.598	0.098
Panel B: Foreign Exchange (5 pairs, 2009–2026)							
AUDUSD	4,241	0.48	0.35	11.3	232	0.577	0.138
EURUSD	4,241	0.28	0.21	5.0	51	0.570	0.263
GBPUSD	4,242	0.32	0.23	27.5	1,117	0.484	0.061
USDCAD	4,240	0.25	0.18	4.3	35	0.654	0.368
USDJPY	4,239	0.33	0.21	14.4	349	0.325	0.063
Panel C: Futures (5 contracts, 2009–2026)							
Corn (C)	4,185	2.48	1.54	12.3	238	0.178	0.079
Crude Oil (CL)	4,223	6.87	2.97	57.6	3,546	0.195	0.047
E-mini S&P (ES)	4,224	1.11	0.53	12.9	212	0.779	0.110
Gold (GC)	4,206	3.63	0.66	62.3	3,963	−0.001	−0.000
Natural Gas (NG)	4,221	10.30	6.63	19.0	522	0.187	0.267

4 Methodology

This section describes the fifteen forecasting models in our comparison: six econometric benchmarks (Section 4.1), nine TSFMs (Section 4.2), and the evaluation framework (Section 4.4).

4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

HAR Model (Corsi, 2009). The HAR model captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$\widehat{RV}_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t}, \tag{1}$$

where $RV_{t-k:t} \equiv k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous k days. We use the original specification, where the weekly and monthly components include overlapping lags (i.e., the weekly component averages days t through $t-4$, not the non-overlapping "rotated" version that separates lags 2–5 from 6–22). This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants operating at different time horizons.

HAR-J Model (Andersen et al., 2007). The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation:

$$\widehat{RV}_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (BPV), a jump-robust estimator of integrated variance constructed from products of adjacent absolute returns (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on J_t indicates that large jumps reduce, rather than increase, future volatility.

HAR-RS Model (Patton and Sheppard, 2015). The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses:

$$\widehat{RV}_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^-, \tag{3}$$

where RS_t^+ and RS_t^- denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure allows upside and downside risk to follow separate dynamics at each aggregation frequency.

HARQ Model (Bollerslev et al., 2016). The HARQ model accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$\widehat{RV}_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t}, \tag{4}$$

where RQ_t is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of RQ_t.

Log-HAR Model (Corsi, 2009). The Log-HAR model applies the HAR specification to log-transformed realized volatility:

$$\widehat{\log RV}_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t}. \tag{5}$$

The logarithmic transformation brings the RV distribution closer to normality, improving ordinary least squares (OLS) efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\widehat{\log RV}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance, following the standard log-normal adjustment.

ARFIMA Model (Granger and Joyeux, 1980; Hosking, 1981). The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing:

$$(1-L)^d\, \Phi(L)\, (\log RV_t - \mu) = \Theta(L)\, \varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter governing the rate at which autocorrelations decay (values near 0.4, typical for RV, imply slow hyperbolic decay rather than the exponential decay of standard ARMA models), $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders p and q, and L is the lag operator. In our implementation, we estimate d via the Geweke–Porter-Hudak (GPH) log-periodogram regression (Geweke and Porter-Hudak, 1983) as a diagnostic measure of long-memory strength, and then fit an ARMA(2,0,2) model directly on log-transformed realized volatility. This two-step approach avoids the numerical instability of truncated fractional differencing when d is near 0.5, while the ARMA(2,2) specification on log(RV) captures short-to-medium-range dynamics adequately for forecasting horizons of 1–22 days.

The log transformation applied in Log-HAR can equally be applied to the HAR-J, HAR-RS, and HARQ specifications, yielding log-space variants that share the same positivity guarantee and improved distributional properties. We focus on Log-HAR as the representative log-space benchmark because it is the most widely used in the literature and because the additional regressors in the extended models (jumps, semivariances, quarticity interactions) operate on the same realized measures regardless of the transformation.

All HAR-family models are estimated by OLS with Newey–West (Newey and West, 1987) heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness. We do not include the Realized GARCH model of Hansen et al. (2012), which jointly models daily returns and realized measures via maximum likelihood. VOLARE provides realized measures but not the daily close-to-close returns required for the return equation, and sourcing returns from an external provider would introduce alignment issues between the two data sources. This omission is unlikely to affect our main conclusions: Hansen et al. (2012) report that Realized GARCH improves upon standard GARCH but does not consistently dominate HAR-family models in out-of-sample forecasting.

4.2 Time Series Foundation Models

We evaluate nine time series foundation models (TSFMs) in a zero-shot setting, applied directly to realized volatility series without any domain-specific training or fine-tuning. The nine models span eight distinct architectures: Chronos-Bolt (small and base checkpoints), Moirai 2.0, Moirai-MoE, Lag-Llama, TimesFM 2.5, Toto, Sundial, and TTM. The zero-shot approach is the most practically relevant deployment mode: it requires no labeled financial data, no model retraining, and can be applied immediately to any new asset. We leave fine-tuning strategies for future work.[4]

Formal framework. A TSFM can be expressed as a parametric mapping from an observed history to a forecast distribution. Let $RV_{1:T} = (RV_1, \ldots, RV_T)$ denote the context window of T past observations (in our case, $T = 512$ daily RV values). A TSFM with pretrained parameters $\hat{\theta}$ produces a forecast of the next H values:

$$\widehat{RV}_{T+1:T+H} = f_{\hat{\theta}}(RV_{1:T}), \tag{7}$$

where $f_{\hat{\theta}}$ maps the input sequence to a predictive distribution over future values.[5] The parameters $\hat{\theta}$ are learned during a pretraining phase on large external corpora of time series spanning weather, energy, retail, transport, and macroeconomic domains. At inference time, the model receives only the context window $RV_{1:T}$ and produces forecasts without any task-specific parameter updates; this is the zero-shot setting.

TSFM components. Every TSFM implements the mapping in Equation (7) through three stages: *tokenization*, *encoding*, and *decoding*. First, the raw series $RV_{1:T}$ is converted

[4] Carriero et al. (2024) additionally evaluate TimeGPT (Garza et al., 2024) for macroeconomic forecasting. We exclude TimeGPT because it is a proprietary, closed-source API that does not permit inspection of model weights or training data, precluding reproducibility.

[5] More precisely, $f_{\hat{\theta}}$ outputs a distribution $\hat{p}(RV_{T+1:T+H} \mid RV_{1:T}; \hat{\theta})$ from which we extract the median as the point forecast. The parameters $\hat{\theta}$ are fixed at their pretrained values throughout our evaluation; no gradient updates occur during inference.

into a sequence of discrete or continuous *tokens*, the model's internal representation of the data. Tokenization serves the same role as variable construction in an econometric model: it determines which features of the history the model can "see." The TSFMs we evaluate use different tokenization strategies (quantile binning, patch projection, lag features, flow matching), summarized in Table 2 and described individually below. Second, the token sequence is processed by a *transformer* (Vaswani et al., 2017), a neural network in which each position attends to other positions via learned attention weights, capturing dependencies at arbitrary lags without imposing the specific functional form that HAR prescribes (daily, weekly, monthly components). Transformers come in two main variants: *encoder-only* models, which read the entire input at once and produce all outputs simultaneously (analogous to a regression that conditions on the full history), and *decoder-only* models, which process the sequence left-to-right and generate outputs one step at a time (analogous to an autoregressive model). Third, the transformer's output is passed through a *decoding head* that maps internal representations back to forecasts in the original scale. Decoding heads differ across models: some output quantiles of the predictive distribution directly, others parameterize an explicit probability distribution (e.g., Student-t) from which forecast trajectories are sampled, and one (Sundial) generates forecast trajectories via learned continuous normalizing flows. In all cases, we extract the median as the point forecast, analogous to reporting the conditional median from a quantile regression.

Table 2: Summary of TSFM architectures evaluated. All models are applied zero-shot with a 512-day context window. "Params" refers to the total number of pretrained parameters. The point forecast is the median of each model's predictive distribution.

Model	Architecture	Tokenization	Params	Output	Release
Chronos-Bolt-S	Encoder-only	Quantile binning	48M	Direct quantiles	2025
Chronos-Bolt-B	Encoder-only	Quantile binning	205M	Direct quantiles	2025
Moirai 2.0-S	Decoder-only	Patch projection	14M	9 quantile levels	2025
Moirai-MoE-S	Decoder-only	Patch projection	43M	9 quantile levels	2024
Lag-Llama	Decoder-only	Lag features	8M	Student-t mixture	2024
TimesFM 2.5	Decoder-only	Patch + residual	200M	Mean + 9 quantiles	2024
Toto	Decoder-only	Patch projection	162M	Student-t mixture	2024
Sundial	Decoder-only	Flow matching	128M	Generative samples	2025
TTM	MLP-Mixer*	Patch + freq. prefix	<1M	Point forecast	2024

*TSMixer variant; see Ekambaram et al. (2024).

Chronos-Bolt (Ansari et al., 2025). Chronos-Bolt is a distilled variant of the Chronos family. The original Chronos model tokenizes continuous values by mapping them into 4,096 discrete bins via quantile binning, where each observation is replaced by the index of its nearest quantile in a reference distribution, much as one would discretize a continuous variable into histogram bins. The model then processes these token indices through a T5 encoder-decoder transformer and generates forecasts autoregressively, predicting one token at a time. Chronos-Bolt replaces this two-step architecture with an encoder-only transformer that reads the entire input context and produces all forecast quantiles in a single forward pass, without iterative decoding. This yields a 250× speedup over the original while maintaining competitive accuracy. The decoding head outputs quantile forecasts directly (e.g., the 10th, 50th, and 90th percentiles), bypassing the need to sample from a token distribution. We use both the small (48M parameters) and base (205M parameters) checkpoints[6] and take the median (0.5 quantile) as the point forecast.

[6] `amazon/chronos-bolt-small` and `amazon/chronos-bolt-base` on HuggingFace. The small and base checkpoints differ only in the number of transformer layers and attention heads; they share the same architecture, tokenization, and training procedure.

Moirai 2.0 (Liu et al., 2025a). Moirai 2.0 builds on the original Moirai architecture. It tokenizes the input series by dividing it into fixed-length *patches* (contiguous blocks of consecutive observations, e.g., 32 days each) and projects each patch into the transformer's embedding space via a learned linear map. This is analogous to constructing non-overlapping moving-window summaries of the series, except that the summary is learned rather than hand-specified. The transformer is decoder-only: it processes patches causally (each patch attends only to itself and earlier patches), preserving the temporal ordering. Moirai 2.0 uses *multi-token prediction*, generating multiple future patches in a single pass rather than one step at a time. The decoding head outputs nine quantile levels (0.1, 0.2, . . . , 0.9) per forecast step. We use the small checkpoint[7] and extract the median quantile as the point forecast.

Moirai-MoE (Liu et al., 2024a). Moirai-MoE extends the Moirai architecture with a sparse Mixture of Experts (MoE) layer that routes each input patch to a subset of specialized expert networks. This allows the model to increase capacity without proportionally increasing inference cost: only a fraction of parameters are active for each input. The small checkpoint has 43M total parameters. Like Moirai 2.0, it uses patch-based tokenization and outputs 9 quantile levels. We use the Moirai-MoE checkpoint[8] and extract the median quantile as the point forecast.

Lag-Llama (Rasul et al., 2024). Lag-Llama is a decoder-only foundation model based on the LLaMA architecture, adapted for univariate probabilistic time series forecasting. Unlike the patch-based models above, Lag-Llama tokenizes each time step individually using a set of *lag features*: at each position t, the input vector consists of the current value and a fixed set of lagged values $(y_{t-1}, y_{t-2}, y_{t-7}, y_{t-14}, \ldots)$, concatenated with time-related covariates. This lag-based design is conceptually closer to an autoregressive econometric specification, since the model explicitly receives lagged values as inputs, though the trans-

[7] `Salesforce/moirai-2.0-R-small` on HuggingFace.
[8] `Salesforce/moirai-moe-1.0-R-small` on HuggingFace.

former learns the mapping from lags to forecasts nonparametrically rather than through a linear projection. The decoding head parameterizes a Student-t mixture distribution at each forecast step: it outputs the location, scale, and degrees-of-freedom parameters for each mixture component. Forecasts are generated by sampling trajectories from this distribution and taking the median across samples. We use the pretrained checkpoint[9] (8 layers, 4 attention heads, 36-dimensional embeddings per head), sample 100 trajectories, and report the median as the point forecast.

TimesFM 2.5 (Das et al., 2024). TimesFM 2.5 is Google's decoder-only foundation model for time series forecasting. It uses a patch-based tokenization with residual connections: the input series is divided into fixed-length patches, and each patch embedding includes a residual connection to the raw input values. The model has 200M parameters and outputs a mean forecast plus quantile levels (0.1, 0.2, . . . , 0.9) per forecast step. TimesFM was pretrained on a mixture of Google Trends data and public time series corpora. We use the TimesFM 2.5 checkpoint[10] and extract the median quantile as the point forecast.

Toto (Cohen et al., 2024). Toto is Datadog's decoder-only foundation model, designed for observability and infrastructure time series. It has 162M parameters and uses patch-based tokenization with a learned linear projection. The decoding head parameterizes a Student-t mixture distribution, outputting location, scale, and degrees-of-freedom parameters for each mixture component. Toto was pretrained primarily on observability metrics (CPU usage, request latency, error rates), making it the only TSFM in our study whose training domain is furthest from financial time series. We use the Toto checkpoint.[11]

Sundial (Liu et al., 2025b). Sundial is a generative foundation model that uses flow matching rather than quantile regression for probabilistic forecasting. Flow matching learns

[9] `time-series-foundation-models/Lag-Llama` on HuggingFace.
[10] `google/timesfm-2.5-200m-pytorch` on HuggingFace.
[11] `Datadog/Toto-Open-Base-1.0` on HuggingFace.

a continuous transformation from a simple noise distribution (e.g., Gaussian) to the forecast distribution, analogous to how a copula maps uniform marginals to a joint distribution but operating in the time-series trajectory space. The model has 128M parameters and generates forecast trajectories by applying this learned transformation. This generative approach produces full distributional forecasts without discretizing the output space into quantile bins. We use the base checkpoint,[12] sample 20 trajectories, and report the median as the point forecast.

TTM (Ekambaram et al., 2024). Tiny Time Mixers (TTM) is IBM's lightweight foundation model based on the TSMixer architecture, a multi-layer perceptron that mixes information across both time steps and channels without using attention. TTM uses frequency prefix tuning: a learned embedding vector indicating the data frequency (e.g., daily, hourly) is prepended to the input, allowing a single pretrained model to handle multiple resolutions. The r2.1 release added daily-frequency support with context lengths ranging from 52 to 512 days. TTM is the smallest model in our evaluation, with fewer than 1M parameters, and runs efficiently on CPU. We use the r2.1 release checkpoint[13] with a 512-day context window and extract the point forecast directly from the model output.

Zero-shot protocol. All TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. The 512-day context length is the maximum supported by all architectures' positional encodings and is the default recommended setting.[14] It corresponds to approximately two years of trading data, providing enough history to capture the long-memory dynamics of realized volatility. By contrast, the

[12] `thuml/sundial-base-128m` on HuggingFace.

[13] `ibm-granite/granite-timeseries-ttm-r2`, branch `512-96-ft-r2.1` on HuggingFace. Here "ft" denotes frequency tuning (not fine-tuning); the model is fully zero-shot.

[14] The 512-day default is common across TSFMs for several reasons: it is a power of two (2^9), which aligns efficiently with GPU memory and yields clean patch counts (e.g., $512/16 = 32$ patches); learned positional encodings are initialized at a fixed maximum length during training, making extrapolation beyond that length unreliable; and 512 is long enough to capture long-memory dynamics yet short enough to accommodate most series in heterogeneous pretraining corpora without excessive padding. Some architectures support longer contexts (Chronos-Bolt up to 2,048), but 512 remains the most heavily validated setting.

252-day rolling estimation window for econometric models follows the convention of using one year of daily data, which is standard in the HAR literature. The one-year window balances the need for sufficient data to estimate HAR's four parameters reliably against the risk of including stale observations from a different volatility regime. This reflects each approach's recommended configuration rather than an equal-information comparison; we view this as the practically relevant setup, since a practitioner would deploy each model with its standard settings. We assess sensitivity to the context window length in Section 7.

For multi-step horizons ($h > 1$), the TSFM produces a trajectory of h individual-step forecasts $(\widehat{RV}_{T+1}, \ldots, \widehat{RV}_{T+h})$, each being the median of the model's predictive distribution at that step. The point forecast is the average of these h values, $h^{-1} \sum_{i=1}^{h} \widehat{RV}_{T+i}$, matching the h-day average RV target defined in Section 4.4. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot. We extract the median rather than the mean of each model's predictive distribution because the median is the optimal point forecast under MAE loss and is more robust to the right-skewed forecast distributions that arise in volatility applications; for symmetric distributions, the two coincide. We acknowledge that the median is not the optimal point forecast under QLIKE loss (the conditional mean is); however, in practice, the median is the standard output of probabilistic TSFMs and produces more stable forecasts than the sample mean when the predictive distribution is heavy-tailed. Table 3 reports wall-clock timing for all models. Econometric specifications require < 0.01 s per forecast step, run entirely on CPU, and impose no special hardware requirements; they execute on commodity hardware including a standard laptop. TSFM inference was conducted on the Duke Compute Cluster (DCC) using a single NVIDIA RTX A5000 GPU (24 GB VRAM), with per-step times ranging from 0.014 s (Chronos-Bolt-Small) to 0.084 s (Lag-Llama). Even the slowest TSFM produces a forecast in under 0.1 s per asset, making the per-step cost negligible relative to the estimation overhead that econometric models avoid entirely in the zero-shot setting.

Table 3: Computational cost per forecast step. TSFM inference was run on the Duke Compute Cluster (DCC) using SLURM job scheduling with a single NVIDIA RTX A5000 GPU (24 GB). Econometric models require only CPU and run on commodity hardware (including a standard laptop) in under one minute per asset; their timings here were measured on a DCC node (Intel Xeon Gold 5320, 2.20 GHz). Seconds per step is the wall-clock time to produce one forecast for one asset-day, averaged across the equity sample ($n \approx 2{,}274$ out-of-sample steps). Total time is the approximate wall-clock for one asset across three horizons. ARFIMA is re-estimated every 22 steps. Lag-Llama timing includes 100 sample paths per step.

Model	Parameters	Device	Sec/step (h=1)	Total (1 asset)
Econometric models				
HAR	4	CPU	0.003	21 s
HAR-J	5	CPU	0.003	21 s
HAR-RS	7	CPU	0.004	27 s
HARQ	5	CPU	0.004	24 s
Log-HAR	4	CPU	0.004	30 s
ARFIMA	≤7	CPU	0.006	45 s
Time series foundation models				
Chronos-Bolt-S	48M	GPU	0.014	1.6 min
Chronos-Bolt-B	205M	GPU	0.025	2.8 min
Moirai-2.0-S	14M	GPU	0.019	2.2 min
Moirai-MoE-S	43M	GPU	0.022	2.5 min
Lag-Llama	8M	GPU	0.084	9.5 min
TimesFM-2.5	200M	GPU	0.031	3.5 min
Toto	162M	GPU	0.028	3.2 min
Sundial	128M	GPU	0.045	5.1 min
TTM	<1M	CPU	0.005	0.6 min

4.3 Pretraining Data and Contamination Risk

A natural concern with zero-shot evaluation is whether TSFM performance is inflated because the models' pretraining corpora overlap with the test series. Table 4 summarizes each model's training data and its financial content. No TSFM in our study was trained on realized volatility. VOLARE's realized variance series are second-moment statistics derived from ultra-high-frequency intraday returns at 5-minute sampling (Cipollini et al., 2026); this quantity does not appear in any known training corpus.

Table 4: TSFM pretraining corpora and financial data exposure. "Financial Share" reports the fraction of total training observations from financial or economic domains. No model was trained on realized volatility or any intraday-derived statistic.

Model	Corpus Size	Fin. Share	Financial Datasets	RV?
Chronos-Bolt	890K series, 84B obs	<0.3%	`exchange_rate`, FRED-MD	No
Moirai 2.0	27.6B obs (LOTSA)	0.10%	Undisclosed	No
Moirai-MoE	27.6B obs (LOTSA)	0.10%	Undisclosed	No
Lag-Llama	8K series	Small	`exchange_rate`	No
TimesFM 2.5	100B+ obs	<1%	Google Trends	No
Toto	Observability data	<0.1%	None (infrastructure metrics)	No
Sundial	Public TS corpora	<1%	Undisclosed	No
TTM	Monash + LOTSA subsets	<1%	`exchange_rate` (via Monash)	No

Financial data constitutes less than 1% of training observations for TSFMs with disclosed corpora (Table 4), limited to daily exchange rates and macroeconomic indicators. For models with undisclosed training data (Moirai 2.0, Sundial, Moirai-MoE), the financial share cannot be precisely quantified, though none report training on realized volatility or intraday-derived statistics. Moreover, most public training corpora (e.g., Monash[15], LOTSA[16]) use time series that end before or during the early portion of our out-of-sample evaluation period (2017–2026), reducing the scope for temporal overlap. This minimal financial exposure makes data contamination an unlikely explanation for TSFM performance on realized volatility. Instead, the results suggest that TSFM accuracy reflects transfer of general temporal patterns (mean reversion, long memory, regime persistence) rather than memorization of financial dynamics.

4.4 Forecast Evaluation

Walk-forward design. We employ a walk-forward evaluation scheme with a rolling origin. Econometric models use a fixed estimation window of 252 trading days (approximately one year): at each origin, the model is estimated on the most recent 252 observations, and the window slides forward one day at a time, producing daily re-estimated forecasts. Foundation

[15] https://forecastingdata.org/

[16] Large-scale Open Time Series Archive; see Woo et al. (2024).

models follow the same walk-forward timing but use a 512-day context window and require no estimation step; only the context window slides forward. Because the TSFM context window is longer, the first TSFM forecast is produced 260 days later than the first econometric forecast; all model comparisons are conducted on the common out-of-sample period where both econometric and TSFM forecasts are available. For the equity sample (2,786 days), this produces approximately 2,148 common out-of-sample forecasts per model–asset combination at $h = 1$; the FX and futures samples (4,240+ days) yield approximately 3,602.

Forecast horizons. We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon h is the average realized volatility over the next h days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct h-step forecasts. For HAR-family models, this means estimating a separate regression for each horizon, where the dependent variable is the h-day average of future RV; the model does not iterate one-step-ahead predictions forward. ARFIMA similarly produces direct multi-step forecasts from its estimated ARMA representation on log RV. For TSFMs, each model generates a full forecast trajectory of length h in a single inference call; the point forecast is the average of the h individual-step medians, matching the h-day average target. This direct approach avoids the compounding of errors that arises in iterated multi-step forecasting.

Loss functions and statistical testing. Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\widehat{RV}_t, RV_t) = \frac{RV_t}{\widehat{RV}_t} - \log \frac{RV_t}{\widehat{RV}_t} - 1, \tag{8}$$

where $\widehat{RV}_t$ is the forecast and RV_t is the realized value. QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE) and mean absolute error (MAE). All forecasts are floored at 10^{-10} to prevent numerical instability in

the QLIKE calculation when models produce near-zero predictions.

Pairwise forecast comparisons use the DM test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating fifteen models, we compute the MCS, which identifies the subset of models whose forecasting ability cannot be statistically distinguished from the best model at a given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

We assess forecast efficiency using the Mincer–Zarnowitz (MZ) regression (Mincer and Zarnowitz, 1969):

$$RV_t = \alpha + \beta \widehat{RV}_t + \varepsilon_t, \tag{9}$$

where $\widehat{RV}_t$ is the model's forecast. Under forecast optimality, $\alpha = 0$ and $\beta = 1$, meaning the forecast is unbiased and captures the correct scale of variation. We test the joint null $H_0 \colon \alpha = 0, \beta = 1$ using a Wald test with Newey–West standard errors and report cross-asset averages and rejection rates at the 5% level.

To examine whether relative forecast performance is stable over time, we apply the Giacomini and Rossi (2010) fluctuation test (GR). For each model paired against the HAR benchmark, we compute a rolling DM statistic over a window of size $m = \lfloor 0.3 \times T \rfloor$, producing a time path of relative QLIKE performance. The test statistic is $\sup_t |S_t|$, compared against critical values from the distribution of the supremum of a standardized Brownian bridge. Rejection of the null indicates that the two models' relative performance changes significantly over time.

5 Empirical Results

This section presents the realized variance forecasting results.[17] We first examine aggregate forecast accuracy across loss functions and horizons, then analyze cross-asset and cross-

[17] Replication code is available at https://github.com/Alessiobrini/human-x-ai-finance.

market heterogeneity.

5.1 Aggregate Forecast Accuracy

This subsection reports the overall forecasting performance averaged across assets within each asset class, without examining individual asset-level variation. The goal is to establish the broad ranking of models by loss function and horizon before turning to cross-asset heterogeneity and statistical testing in subsequent subsections.

Table 5 reports the cross-sectional average of per-asset loss functions across 40 equities. Figure 1 illustrates the forecasts of three representative models against realized values for four assets spanning equities and FX. The results reveal a systematic divergence between QLIKE and the other metrics.

Table 5: Forecast accuracy for 40 U.S. equities (VOLARE). Values are cross-sectional averages of per-asset loss functions. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE
Panel: $h = 1$ (1 day)			
HAR	0.243	1.42	1.38†
HAR-J	0.241	1.41	1.11†
HAR-RS	0.264	1.51	3.63†
HARQ	0.296	1.49	16.04†
Log-HAR	0.255	1.31	0.219
ARFIMA	0.300	1.63	0.322
Chronos-Bolt-S	0.216	1.18	0.229
Chronos-Bolt-B	0.213	1.17	0.230
Moirai-2.0-S	0.210	1.19	0.226
Moirai-MoE-S	0.214	1.23	0.211
Lag-Llama	0.291	1.65	0.402
TimesFM-2.5	**0.208**	**1.16**	0.222
Toto	0.301	2.41	34.59†
Sundial	0.210	1.21	**0.195**
TTM	0.217	1.88	0.314
Panel: $h = 5$ (5 days)			
HAR	0.161	1.39	2.70†
HAR-J	0.158	1.38	2.47†
HAR-RS	0.183	1.49	6.83†
HARQ	0.190	1.42	15.89†
Log-HAR	0.161	1.26	0.229
ARFIMA	0.175	1.44	0.251
Chronos-Bolt-S	0.128	1.09	0.213
Chronos-Bolt-B	0.128	1.09	0.216
Moirai-2.0-S	0.122	1.07	0.188
Moirai-MoE-S	0.122	1.08	0.167
Lag-Llama	0.181	1.43	0.305
TimesFM-2.5	0.122	**1.06**	0.207
Toto	0.169	1.67	16.14†
Sundial	**0.120**	1.09	**0.161**
TTM	**0.120**	1.60	0.227
Panel: $h = 22$ (22 days)			
HAR	0.116	1.46	5.92†
HAR-J	0.115	1.45	5.79†
HAR-RS	0.128	1.55	9.28†
HARQ	0.130	1.47	12.02†
Log-HAR	0.113	1.34	0.345
ARFIMA	0.123	1.34	0.280
Chronos-Bolt-S	0.101	1.13	0.322
Chronos-Bolt-B	0.101	1.14	0.325
Moirai-2.0-S	0.098	**1.07**	0.257
Moirai-MoE-S	0.139	1.14	0.233
Lag-Llama	0.109	1.25	0.283
TimesFM-2.5	0.099	1.11	0.326
Toto	0.281	1.61	6.44†
Sundial	0.092	1.08	0.218
TTM	**0.090**	1.26	**0.190**

† Marks QLIKE > 1. For HAR-RS and HARQ, this reflects near-zero forecasts from levels-based OLS.

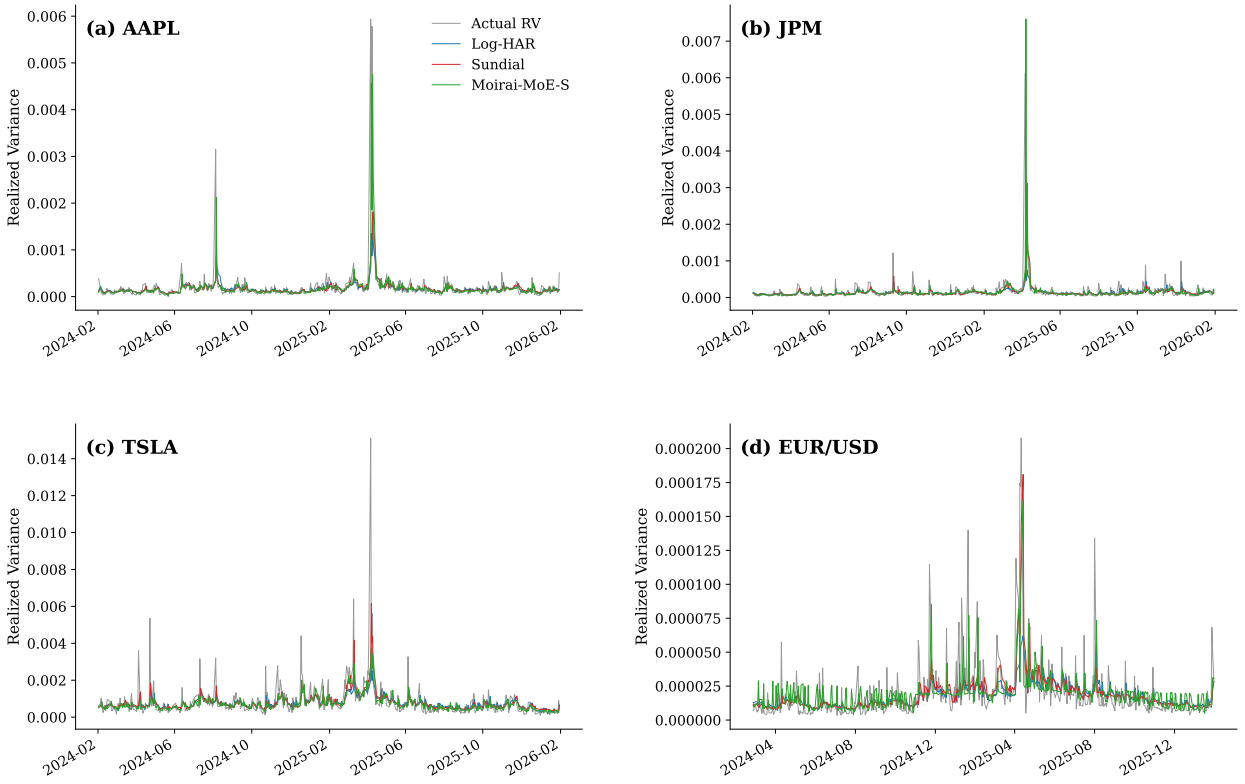

Figure 1: Forecast vs. actual realized variance at $h = 1$ for four representative assets (AAPL, JPM, TSLA, EUR/USD). The last 500 out-of-sample observations are shown. Gray: actual RV; blue: Log-HAR; red: Sundial; green: Moirai-MoE-S.

At the one-day horizon ($h = 1$), Sundial achieves the lowest average QLIKE (0.195), followed by Moirai-MoE-S (0.211), Log-HAR (0.219), and TimesFM 2.5 (0.222). The gap between Sundial and the best econometric model (Log-HAR) is 11% in relative terms, a substantial margin for a zero-shot model with no financial training data. On MSE and MAE, the top TSFMs again lead: TimesFM 2.5 achieves the lowest MSE (0.208×10^{-6}) and lowest MAE (1.16×10^{-4}), while Sundial and Moirai 2.0 follow closely on MSE (0.210×10^{-6}). However, not all TSFMs perform well: Toto produces severely miscalibrated forecasts (QLIKE = 34.59, MSE = 0.301×10^{-6}), consistent with its pretraining on observability metrics (CPU usage, request latency) that share little distributional overlap with realized volatility. However, MZ bias correction reveals that Toto's forecasts contain signal beneath

the scale misalignment (Section 7). TTM achieves a competitive MSE (0.217×10^{-6}) but lags on QLIKE (0.314) and MAE (1.88×10^{-4}), indicating that its MLP-Mixer architecture captures directional variation at the daily horizon but with less precise calibration than the top TSFMs. Lag-Llama (QLIKE = 0.402) underperforms Log-HAR and ARFIMA but outperforms the levels-based HAR variants whose QLIKE is inflated by near-zero forecasts.

At the five-day horizon ($h = 5$), the TSFM advantage becomes more pronounced. Sundial achieves the best QLIKE (0.161) and MSE (0.120×10^{-6}), with Moirai-MoE-S close behind (QLIKE = 0.167). TimesFM 2.5 achieves the lowest MAE (1.06×10^{-4}) and ties Moirai 2.0 and Moirai-MoE-S for second-lowest MSE (0.122×10^{-6}). TTM ties Sundial for the lowest MSE (0.120×10^{-6}) but its QLIKE (0.227) and MAE (1.60×10^{-4}) remain weaker, suggesting that its point forecasts track the level of volatility well in MSE terms but are less precisely calibrated under asymmetric loss. The gap over Log-HAR in QLIKE terms is now 30% for Sundial, compared to 11% at $h = 1$. The econometric models that do best at $h = 1$ (HAR, HAR-J) deteriorate more rapidly at longer horizons, while the top TSFMs maintain relatively stable accuracy.

At the monthly horizon ($h = 22$), TTM achieves the lowest equity QLIKE (0.190) and lowest MSE (0.090×10^{-6}), surpassing Sundial (QLIKE = 0.218, MSE = 0.092×10^{-6}). This is a striking result: TTM is the smallest model in the evaluation (<1M parameters), yet it outperforms all other models at the monthly horizon for equities. Sundial achieves the second-lowest QLIKE, followed by Moirai-MoE-S (0.233) and Moirai 2.0 (0.257), while Moirai 2.0 achieves the lowest MAE (1.07×10^{-4}). The econometric benchmarks show substantially higher MSE at this horizon (e.g., Log-HAR: 0.113×10^{-6} vs. TTM: 0.090×10^{-6}), indicating that HAR-family models lose accuracy at the monthly horizon relative to the best TSFMs. This horizon-dependent pattern is consistent with the hypothesis that TSFMs' large context windows (512 days) capture long-range dependencies more effectively than the HAR's fixed three-component structure. TTM's improvement from weak performance at $h = 1$ (QLIKE = 0.314) to best-in-class at $h = 22$ (QLIKE = 0.190) suggests that

its MLP-Mixer architecture, which mixes information across time steps without attention, is particularly effective at capturing the slow mean reversion that dominates the monthly horizon. The sub-sample analysis in Section 7 confirms that these rankings hold in both the pre-COVID and post-COVID periods.

Cross-sectional medians (Table 18 in Appendix A) also confirm that these rankings are robust to outlier assets: under median aggregation, Sundial achieves the best median QLIKE at $h = 1$ and $h = 5$, while TTM achieves the best median QLIKE at $h = 22$ (0.186 vs. Sundial's 0.217), and the gap between TSFMs and econometric models narrows only modestly.

Levels-based HAR extensions are unstable in this evaluation. Unconstrained OLS estimation of HAR variants can produce negative or near-zero forecasts. HAR-RS and HARQ produce QLIKE values of 3.63 and 16.04 at $h = 1$, respectively, moving to 9.28 and 12.02 at $h = 22$. These inflated values arise because the OLS-estimated models occasionally produce near-zero forecasts for RV_{t+h}, which are floored at 10^{-10} for QLIKE evaluation. Across the full sample, no forecast is literally negative, but the floor binds for a small fraction of observations. Since QLIKE $= RV/\hat{RV} - \log(RV/\hat{RV}) - 1$, a forecast near zero generates an extremely large loss whenever the realized value is far from zero, and a handful of such observations can dominate the cross-sectional average. Across the equity sample, the 10^{-10} floor binds for fewer than 0.1% of TSFM forecast–date pairs but for 2–5% of HAR-RS and HARQ pairs, explaining their inflated QLIKE. The median results in Table 18 confirm that the qualitative ranking of models is unchanged. Log-HAR and ARFIMA avoid this problem entirely by operating in log space, where forecasts are positive by construction. TSFMs also produce strictly positive forecasts in our evaluation, since their median output on a strictly positive input series does not cross zero. This practical advantage may be as important as any differences in statistical accuracy: a model that occasionally produces near-zero forecasts is difficult to deploy in production risk systems.

5.2 Cross-Asset and Cross-Market Heterogeneity

The equity results above establish a clear ranking, but it is not obvious that these patterns generalize to asset classes with different volatility dynamics. Tables 6 and 7 extend the analysis to foreign exchange and futures markets.

Table 6: Forecast accuracy for 5 FX pairs (VOLARE, 2009–2026). Values are cross-sectional averages of per-asset loss functions across five major currency pairs. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

Model	MSE ($\times 10^{-8}$)	MAE ($\times 10^{-4}$)	QLIKE
Panel: $h = 1$ (1 day)			
HAR	0.174	0.14	0.177
HAR-J	0.170	0.14	0.196
HAR-RS	0.192	0.15	0.249
HARQ	0.196	0.14	0.628
Log-HAR	0.174	0.13	0.163
ARFIMA	0.207	0.16	0.242
Chronos-Bolt-S	0.168	0.12	0.174
Chronos-Bolt-B	0.167	0.12	0.176
Moirai-2.0-S	0.222	0.19	0.329
Moirai-MoE-S	0.178	0.14	0.199
Lag-Llama	0.207	0.17	0.305
TimesFM-2.5	0.166	**0.12**	0.171
Toto	3.747	1.21	12.65†
Sundial	**0.162**	0.12	**0.153**
TTM	0.679	0.75	0.809
Panel: $h = 5$ (5 days)			
HAR	0.082	0.12	0.143
HAR-J	0.079	0.12	0.149
HAR-RS	0.098	0.13	0.240
HARQ	0.085	0.12	0.399
Log-HAR	0.080	0.11	0.120
ARFIMA	0.098	0.13	0.178
Chronos-Bolt-S	0.075	0.10	0.128
Chronos-Bolt-B	0.075	0.10	0.129
Moirai-2.0-S	0.110	0.17	0.250
Moirai-MoE-S	0.076	0.11	0.139
Lag-Llama	0.098	0.14	0.194
TimesFM-2.5	0.074	**0.10**	0.126
Toto	0.881	0.62	10.97†
Sundial	**0.068**	0.10	**0.098**
TTM	0.348	0.55	0.588
Panel: $h = 22$ (22 days)			
HAR	0.052	0.13	0.186
HAR-J	0.052	0.12	0.197
HAR-RS	0.065	0.13	0.223
HARQ	0.053	0.12	0.367
Log-HAR	0.049	0.12	0.147
ARFIMA	0.060	0.12	0.185
Chronos-Bolt-S	0.047	0.10	0.161
Chronos-Bolt-B	0.047	0.10	0.163
Moirai-2.0-S	0.064	0.16	0.212
Moirai-MoE-S	0.062	0.11	0.145
Lag-Llama	0.053	0.13	0.164
TimesFM-2.5	0.046	**0.10**	0.165
Toto	0.722	0.37	9.23†
Sundial	**0.042**	0.10	**0.112**
TTM	0.060	0.18	0.188

Table 7: Forecast accuracy for 5 futures contracts (VOLARE, 2009–2026). Values are cross-sectional averages of per-asset loss functions across five contracts. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to near-zero forecasts or systematic forecast bias.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE
Panel: $h = 1$ (1 day)			
HAR	14.818	2.72	0.225
HAR-J	14.958	2.65	0.707
HAR-RS	15.503	2.81	5.82†
HARQ	15.156	2.65	22.34†
Log-HAR	14.143	2.09	0.193
ARFIMA	16.437	2.95	0.346
Chronos-Bolt-S	14.335	1.99	0.213
Chronos-Bolt-B	14.251	1.95	0.221
Moirai-2.0-S	14.509	1.99	0.289
Moirai-MoE-S	21.772	2.19	0.211
Lag-Llama	14.860	2.77	1.12†
TimesFM-2.5	**14.084**	**1.92**	0.204
Toto	15.080	3.23	24.90†
Sundial	14.246	2.05	**0.189**
TTM	14.458	3.00	0.411
Panel: $h = 5$ (5 days)			
HAR	4.574	2.56	0.434
HAR-J	4.918	2.48	0.839
HAR-RS	4.615	2.60	4.07†
HARQ	4.792	2.49	22.10†
Log-HAR	4.442	1.98	0.165
ARFIMA	6.262	2.70	0.269
Chronos-Bolt-S	4.352	1.80	0.185
Chronos-Bolt-B	4.384	1.77	0.196
Moirai-2.0-S	4.523	1.80	0.207
Moirai-MoE-S	4.501	1.76	0.155
Lag-Llama	4.859	2.48	0.333
TimesFM-2.5	4.262	**1.74**	0.175
Toto	4.202	2.35	12.92†
Sundial	**3.977**	1.79	**0.139**
TTM	4.259	2.61	0.259
Panel: $h = 22$ (22 days)			
HAR	2.169	2.77	1.08†
HAR-J	2.234	2.73	1.23†
HAR-RS	2.201	2.89	4.01†
HARQ	2.126	2.66	17.88†
Log-HAR	2.001	2.16	0.280
ARFIMA	3.681	2.65	0.286
Chronos-Bolt-S	1.873	1.92	0.282
Chronos-Bolt-B	1.886	1.87	0.286
Moirai-2.0-S	1.855	1.83	0.238
Moirai-MoE-S	1.625	**1.78**	0.206
Lag-Llama	2.062	2.29	0.336
TimesFM-2.5	1.816	1.85	0.275
Toto	1.598	2.19	6.37†
Sundial	**1.595**	1.84	**0.184**
TTM	1.765	2.30	0.202

Foreign exchange. The FX sample covers five major currency pairs with lower volatility levels ($\bar{RV} \approx 3 \times 10^{-5}$) and moderate persistence ($\bar{\rho}_1 = 0.52$). For FX, Sundial achieves the lowest QLIKE and lowest MSE across all horizons, while TimesFM 2.5 achieves the lowest MAE. At $h = 1$, Sundial achieves the lowest QLIKE (0.153) and MSE (0.162×10^{-8}), followed by Log-HAR (0.163) and TimesFM 2.5 (0.171). Moirai 2.0 underperforms substantially (QLIKE = 0.329, MSE = 0.222×10^{-8}). At $h = 5$, Sundial extends its lead (QLIKE = 0.098 vs. 0.120 for Log-HAR), with TimesFM 2.5 (0.126) and Chronos-Bolt (0.128–0.129) close behind. At $h = 22$, Sundial again leads on QLIKE (0.112) and MSE (0.042×10^{-8}), while Moirai-MoE-S (0.145) outperforms Log-HAR (0.147). TTM performs poorly on FX (QLIKE = 0.809 at $h = 1$, 0.188 at $h = 22$), in sharp contrast to its strong equity results, indicating that its MLP-Mixer architecture does not transfer well to the lower-amplitude FX dynamics. Moirai 2.0's poor FX performance (highest QLIKE among TSFMs at all horizons) is not driven by a single currency pair: per-asset QLIKE analysis (Table 19 in Appendix A) confirms that Moirai underperforms all competing models on every FX pair at every horizon. This systematic weakness stands in contrast to its equity strength and suggests that Moirai's training data or architecture is less suited to the smoother, lower-amplitude dynamics of currency volatility.

Futures. The futures sample displays the widest cross-asset heterogeneity in volatility dynamics. The E-mini S&P 500 (ES) has first-order autocorrelation of 0.779, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$). At $h = 1$, Sundial achieves the lowest QLIKE (0.189), followed by Log-HAR (0.193) and TimesFM 2.5 (0.204). Moirai 2.0 ranks behind (0.289). At $h = 5$, Sundial leads on QLIKE (0.139), followed by Moirai-MoE-S (0.155) and Log-HAR (0.165); Sundial also achieves the lowest MSE (3.977×10^{-6}). At the monthly horizon, Sundial retains the QLIKE lead (0.184), followed by Moirai-MoE-S (0.206), Moirai 2.0 (0.238), and TimesFM 2.5 (0.275), while levels-based HAR variants produce inflated QLIKE (HAR: 1.08, HARQ: 17.88).

Cross-asset patterns. Three patterns emerge from the cross-asset comparison. First, Sundial achieves the lowest QLIKE across all three asset classes and nearly all horizons (Tables 5, 6, 7), with the sole exception of equities at $h = 22$, where TTM achieves a lower QLIKE (0.190 vs. 0.218). This level of consistency is not achieved by any other model in the evaluation. Second, Log-HAR remains the best or near-best econometric specification across all asset classes, confirming the practical value of the log transformation for ensuring forecast positivity and aligning with QLIKE's loss structure. Third, within the TSFM class, performance dispersion is wide: Sundial, Moirai-MoE-S, TimesFM 2.5, Moirai 2.0, and (at longer horizons) TTM form a competitive tier, while Toto underperforms across all asset classes and Lag-Llama underperforms at short horizons. Per-asset QLIKE breakdowns in Appendix A (Tables 19–21) confirm that these rankings are not driven by outlier assets. These results highlight the importance of both multi-market and multi-model evaluations: a model that appears dominant on one asset class may not generalize.

Figure 2 provides a distributional view of these results, plotting QLIKE ratios (model / HAR) across all 50 assets for selected models. Values below one indicate the model outperforms HAR. At $h = 1$, Sundial has the lowest median ratio and the tightest interquartile range, while Lag-Llama's median ratio exceeds 1.7. At $h = 5$, Sundial and Moirai-MoE-S pull ahead of the field, with the vast majority of assets below one. By $h = 22$, all well-calibrated models improve substantially relative to HAR, with Sundial maintaining the lowest median ratio. The box plots also reveal heterogeneity within the TSFM class: Toto's ratio consistently exceeds HAR across all horizons, confirming its poor calibration, while Sundial and Moirai-MoE-S show the tightest distributions, indicating reliable cross-asset performance.

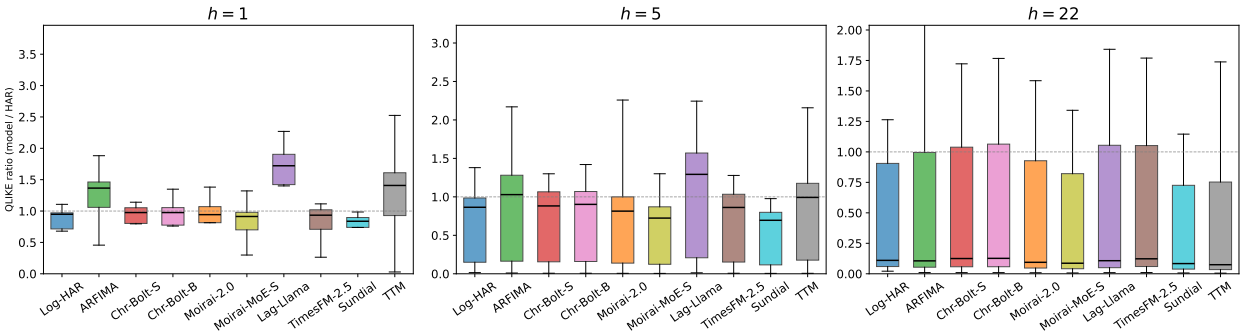

Figure 2: Distribution of QLIKE ratios (model / HAR) across 50 assets for ten selected models. Values below one indicate the model outperforms HAR. Boxes span the interquartile range; whiskers extend to 1.5× IQR; outliers are suppressed. The dashed line marks parity with HAR. HAR, HAR-J, HAR-RS, HARQ, and Toto are excluded because their extreme QLIKE ratios compress the scale for the remaining models.

5.3 Drivers of TSFM Forecast Performance

The cross-asset results above show that TSFM rankings vary across asset classes, raising the question of what asset-level characteristics drive relative forecast performance. Carriero et al. (2024) find that TSFMs struggle with highly persistent macroeconomic series (first-order partial autocorrelation above 0.7). We test whether a similar pattern holds for realized volatility by examining the relationship between each asset's first-order autocorrelation ρ_1 and the TSFM's QLIKE ratio relative to HAR.

Figure 3 plots QLIKE ratios (model / HAR) at $h = 1$ against ρ_1 for 50 assets, separately for each TSFM. Unlike the macroeconomic setting, we find no statistically significant relationship between persistence and TSFM relative performance: the rank correlation ranges from $r = 0.06$ ($p = 0.68$) for Chronos-Bolt to $r = -0.17$ ($p = 0.23$) for Lag-Llama, with the remaining models showing similarly insignificant correlations. Sundial achieves lower QLIKE than HAR for 100% of assets in both high- and low-persistence groups, confirming its consistent advantage regardless of persistence. Moirai-MoE-S beats HAR for 84% of high-persistence and 68% of low-persistence assets, while TimesFM 2.5 shows a modest

persistence effect (76% vs. 56%).

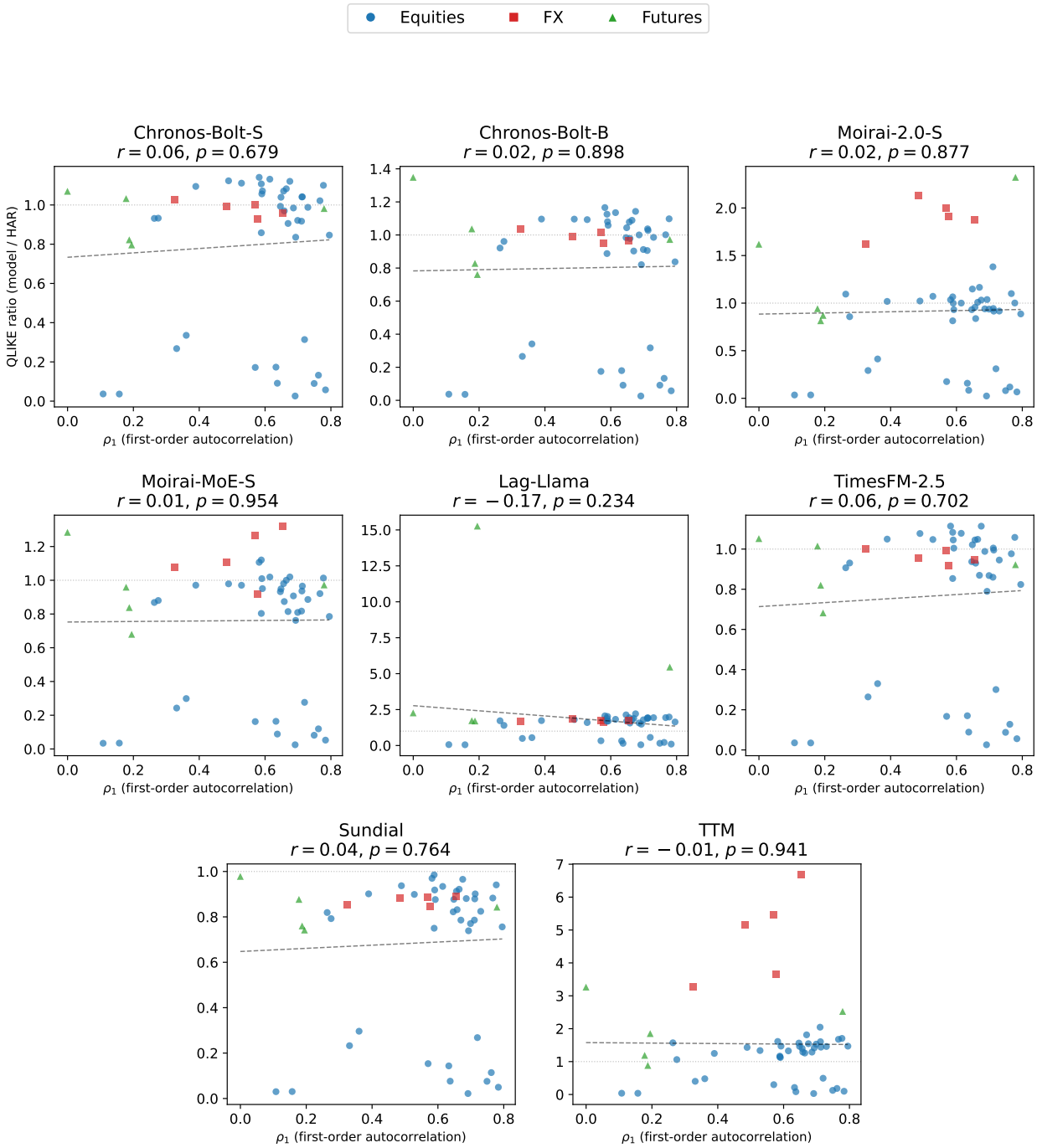

Figure 3: QLIKE ratio (model / HAR) at $h = 1$ vs. first-order autocorrelation ρ_1 for 50 assets. Values below one indicate the TSFM outperforms HAR. Each point is one asset, colored by class (blue: equities, red: FX, green: futures). Dashed line: OLS trend; r and p are the Pearson correlation and its p-value. Toto is omitted due to extreme QLIKE ratios. Eight TSFMs are shown.

This null result is informative (Figure 3). Realized volatility, unlike many macroeconomic series, is stationary and mean-reverting by construction. Even the least persistent assets in our sample (Gold, $\rho_1 \approx 0$) exhibit the positive, right-skewed distribution that TSFMs encounter frequently in their pretraining corpora. The variation in TSFM performance across assets is better explained by architecture than by persistence: Sundial achieves the lowest QLIKE across all asset classes regardless of individual ρ_1, while other TSFMs show asset-class-dependent strengths. This suggests that the relevant driver is the match between each TSFM's implicit prior (shaped by its training data composition and output mechanism) and the distributional characteristics of realized volatility, rather than the degree of serial dependence.

6 Statistical Significance and Model Confidence Sets

The rankings in Section 5 show large differences across models, but mean loss comparisons can be misleading when distributions are skewed and sample sizes vary across assets. We now apply four formal statistical tests: MCS to identify the subset of models that cannot be distinguished from the best, pairwise DM tests to quantify directional win rates, MZ regressions to assess forecast efficiency, and GR fluctuation tests to detect time variation in relative performance.

Table 8 reports MCS inclusion rates for all 50 assets, Figure 4 visualizes the inclusion patterns, and Table 9 summarizes pairwise DM test results across all 50 assets.

Table 8: Model Confidence Set inclusion rates for 50 assets (40 equities, 5 FX, 5 futures). Each cell reports the percentage of assets for which the model is included in the MCS at the 10% significance level (QLIKE loss, $T_{\max}$ statistic, block bootstrap with $B = 10{,}000$).

Model	$h = 1$ (%)	$h = 5$ (%)	$h = 22$ (%)
HAR	4.0	8.0	6.0
HAR-J	0.0	6.0	6.0
HAR-RS	0.0	0.0	6.0
HARQ	0.0	0.0	0.0
Log-HAR	36.0	18.0	46.0
ARFIMA	0.0	4.0	50.0
Chronos-Bolt-S	0.0	0.0	12.0
Chronos-Bolt-B	0.0	0.0	10.0
Moirai-2.0-S	6.0	20.0	52.0
Moirai-MoE-S	20.0	60.0	68.0
Lag-Llama	0.0	0.0	46.0
TimesFM-2.5	2.0	0.0	12.0
Toto	0.0	0.0	0.0
Sundial	**100.0**	**100.0**	**92.0**
TTM	0.0	4.0	86.0

Table 9: Diebold–Mariano test: pairwise win rates across 50 assets (40 equities, 5 FX, 5 futures). Each cell reports the percentage of pairwise comparisons (across 50 assets $\times$ 14 opponents) in which the row model achieves significantly lower QLIKE at the 5% level.

Model	$h = 1$	$h = 5$	$h = 22$
HAR	39.7%	32.8%	25.5%
HAR-J	26.4%	29.3%	24.1%
HAR-RS	18.9%	14.1%	12.1%
HARQ	5.1%	4.8%	4.1%
Log-HAR	61.6%	51.5%	52.3%
ARFIMA	27.9%	29.1%	46.5%
Chronos-Bolt-S	46.7%	44.8%	39.1%
Chronos-Bolt-B	46.4%	41.7%	36.5%
Moirai-2.0-S	51.5%	58.5%	56.5%
Moirai-MoE-S	63.9%	77.5%	67.9%
Lag-Llama	18.1%	22.1%	47.3%
TimesFM-2.5	57.9%	52.4%	37.7%
Toto	0.4%	2.0%	6.0%
Sundial	**88.7%**	**88.9%**	82.9%
TTM	23.7%	31.9%	**83.5%**

Figure 4: Model Confidence Set inclusion for 50 assets. Green cells indicate that the model is included in the MCS at the 10% significance level (QLIKE loss). Left: $h = 1$; center: $h = 5$; right: $h = 22$.

MCS results. At $h = 1$, Sundial enters the MCS for 100% of assets, confirming its status as the single most reliable specification at the daily horizon. Log-HAR enters for 36%, Moirai-MoE-S for 20%, Moirai 2.0 for 6%, HAR for 4%, and TimesFM 2.5 for 2%. Nine of fifteen models enter for 0%, including all levels-based HAR variants except HAR, both Chronos-Bolt checkpoints, Lag-Llama, TTM, and Toto.

At $h = 5$, TSFMs continue to lead. Sundial again enters for 100% of assets, followed by Moirai-MoE-S (60%), Moirai 2.0 (20%), Log-HAR (18%), HAR (8%), HAR-J (6%), TTM (4%), and ARFIMA (4%).

At $h = 22$, the ranking broadens substantially. Sundial enters for 92%, followed by TTM (86%), Moirai-MoE-S (68%), Moirai 2.0 (52%), ARFIMA (50%), Log-HAR and Lag-Llama

(both 46%). TTM's sharp improvement from 0% at $h = 1$ to 86% at $h = 22$ is one of the most notable findings: the smallest model in our evaluation (<1M parameters) enters the MCS for nearly nine out of ten assets at the monthly horizon. Lag-Llama exhibits a similar horizon-dependent pattern, improving from 0% at $h = 1$ to 46% at $h = 22$, suggesting that its lag-based tokenization captures long-range dependencies more effectively than short-term dynamics. The high ARFIMA inclusion rate at $h = 22$ is consistent with its long-memory parameterization: at the monthly horizon, fractional differencing captures slow mean reversion that the HAR's three fixed lags approximate less precisely.

DM pairwise tests. The DM test is computed pairwise: for each of the 15 models reported in Table 9, we test it against each of the remaining 14 on each of the 50 assets, yielding $15 \times 14 \times 50 = 10{,}500$ individual tests. Table 9 reports the fraction of pairwise DM tests in which each model achieves significantly lower QLIKE at the 5% level, aggregated across 50 assets and 14 opponents. At $h = 1$, Sundial wins the highest fraction of pairwise comparisons (88.7%), followed by Moirai-MoE-S (63.9%), Log-HAR (61.6%), and TimesFM 2.5 (57.9%). TTM achieves 23.7% at $h = 1$, consistent with its weaker daily-horizon accuracy. At $h = 5$, Sundial leads (88.9%), followed by Moirai-MoE-S (77.5%) and Moirai 2.0 (58.5%). TTM improves to 31.9%. At $h = 22$, TTM achieves the highest win rate (83.5%), marginally surpassing Sundial (82.9%), with Moirai-MoE-S (67.9%) and Moirai 2.0 (56.5%) following. This reversal at the monthly horizon is consistent with TTM's strong QLIKE performance at $h = 22$. Toto's win rate never exceeds 6%, confirming its persistent miscalibration.

Forecast efficiency: MZ regressions. Table 10 reports average MZ regression results across 50 assets. The MZ regression, $RV_t = \alpha + \beta \widehat{RV}_t + \varepsilon_t$, tests whether forecasts are efficient: under the null, $\alpha = 0$ and $\beta = 1$.

At $h = 1$, Moirai 2.0 and ARFIMA produce the most well-calibrated forecasts ($\hat{\beta} = 1.05$ and 1.04, respectively), closest to the theoretical ideal of unity. Sundial ($\hat{\beta} = 1.14$) and TimesFM 2.5 ($\hat{\beta} = 1.18$) also remain close to unity. The HAR family overshoots modestly

($\hat{\beta} = 1.28$ for HAR, 1.24 for HAR-J), while Log-HAR overshoots substantially ($\hat{\beta} = 1.76$), reflecting the bias-correction step that converts log-space forecasts back to levels. Among TSFMs, TimesFM 2.5 achieves the highest MZ R^2 (0.432), followed by Chronos-Bolt-B (0.423) and Moirai-MoE-S (0.415). Two models show systematic undershooting: Moirai-MoE-S ($\hat{\beta} = 0.79$) and Toto ($\hat{\beta} = 0.44$), indicating compressed forecasts. The joint efficiency null ($\alpha = 0$, $\beta = 1$) is rejected for 90–100% of assets for most TSFMs (76% for Lag-Llama), compared to 54–58% for HAR and HAR-J. This seemingly paradoxical result (TSFMs are more accurate on MSE and R^2 but more frequently reject efficiency) arises because the MZ test has greater power when R^2 is high.

Table 10: Forecast accuracy and Mincer–Zarnowitz efficiency ($h = 1$). Cross-sectional averages across 50 assets (40 equities, 5 FX, 5 futures). The MZ regression is $RV_t = \alpha + \beta\, \widehat{RV}_t + \varepsilon_t$; under forecast efficiency, $\alpha = 0$ and $\beta = 1$. The F-statistic tests the joint null with HAC (Newey–West) standard errors. % Reject reports the fraction of assets for which the joint null is rejected at the 5% level.

	Loss functions			MZ regression				
Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	% Reject
Econometric benchmarks								
HAR	1.677	1.43	1.141	−0.0000	1.278	0.368	20.59	58%
HAR-J	1.688	1.41	0.980	−0.0000	1.239	0.377	10.95	54%
HAR-RS	1.762	1.50	3.515	0.0000	1.118	0.313	23.93	70%
HARQ	1.752	1.47	15.125	0.0001	0.896	0.145	13.47	82%
Log-HAR	1.618	1.27	0.211	−0.0001	1.760	0.314	7.99	86%
ARFIMA	1.883	1.61	0.316	0.0001	1.042	0.167	21.66	84%
Foundation models								
Chronos-Bolt-S	1.607	1.15	0.222	0.0000	1.200	0.402	34.51	98%
Chronos-Bolt-B	1.596	1.14	0.224	0.0000	1.253	0.423	42.74	98%
Moirai-2.0-S	1.619	1.17	0.243	0.0000	1.050	0.378	22.16	98%
Moirai-MoE-S	2.348	1.21	0.209	0.0001	0.789	0.415	17.75	100%
Lag-Llama	1.719	1.62	0.464	−0.0001	1.499	0.147	7.17	76%
TimesFM-2.5	1.575	1.13	0.215	0.0000	1.183	0.432	36.89	98%
Toto	1.752	2.37	31.424	0.0001	0.441	0.257	5860.06	100%
Sundial	1.593	1.19	0.190	0.0000	1.138	0.412	6.67	90%
TTM	1.620	1.88	0.373	−0.0001	0.907	0.403	1781.81	98%

At $h = 5$ (Table 11), the calibration gap between TSFMs and econometric models widens. Among econometric benchmarks, HAR and HAR-J maintain $\hat{\beta}$ close to unity (1.15), but HARQ undershoots ($\hat{\beta} = 0.69$) and Log-HAR overshoots ($\hat{\beta} = 1.45$). ARFIMA's $\hat{\beta}$ drops to 0.96, remaining well calibrated. Among TSFMs, Moirai 2.0 achieves the best calibra-

tion ($\hat{\beta} = 1.09$), while Moirai-MoE-S undershoots more sharply ($\hat{\beta} = 0.75$), indicating compressed forecasts at this horizon. Sundial maintains $\hat{\beta} = 1.14$ with the lowest QLIKE (0.153). TimesFM 2.5 achieves the highest R^2 (0.437), followed by Moirai-MoE-S (0.444). TTM's calibration is near-optimal ($\hat{\beta} = 0.96$) despite its weaker QLIKE (0.266), and its R^2 (0.402) matches or exceeds most TSFMs, indicating that its MLP-Mixer architecture captures predictive variation effectively even at the weekly horizon. Toto's severe undershoot persists ($\hat{\beta} = 0.48$).

Table 11: Forecast accuracy and Mincer–Zarnowitz efficiency ($h = 5$). Cross-sectional averages across 50 assets. The MZ regression is $RV_t = \alpha + \beta\, \widehat{RV}_t + \varepsilon_t$; under forecast efficiency, $\alpha = 0$ and $\beta = 1$. % Reject reports the fraction of assets for which the joint null is rejected at the 5% level (Wald test, Newey–West standard errors).

	Loss functions			MZ regression				
Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	% Reject
Econometric benchmarks								
HAR	0.586	1.38	2.216	0.0000	1.151	0.282	7.05	28%
HAR-J	0.618	1.36	2.073	0.0000	1.150	0.299	5.94	28%
HAR-RS	0.608	1.47	5.898	0.0001	0.908	0.217	15.55	52%
HARQ	0.631	1.40	14.959	0.0001	0.686	0.126	12.93	68%
Log-HAR	0.573	1.22	0.211	−0.0000	1.445	0.209	5.26	72%
ARFIMA	0.766	1.43	0.246	0.0001	0.963	0.215	20.27	72%
Foundation models								
Chronos-Bolt-S	0.537	1.07	0.202	0.0000	1.282	0.400	26.31	98%
Chronos-Bolt-B	0.541	1.06	0.205	0.0000	1.438	0.417	31.25	98%
Moirai-2.0-S	0.550	1.05	0.196	0.0000	1.094	0.367	18.13	94%
Moirai-MoE-S	0.548	1.05	0.163	0.0001	0.751	0.444	16.97	98%
Lag-Llama	0.631	1.40	0.297	0.0000	1.182	0.068	4.99	60%
TimesFM-2.5	0.524	1.03	0.196	0.0000	1.324	0.437	30.22	98%
Toto	0.556	1.63	15.301	0.0001	0.482	0.337	1570.66	98%
Sundial	0.494	1.06	0.153	0.0000	1.136	0.405	5.60	82%
TTM	0.522	1.59	0.266	−0.0001	0.958	0.402	678.22	98%

At $h = 22$ (Table 12), all econometric models exhibit substantial undershoot ($\hat{\beta} = 0.32$–0.57), with MZ R^2 falling to 0.08–0.14. TSFMs retain substantially better calibration: Moirai 2.0 achieves $\hat{\beta} = 0.99$, Sundial $\hat{\beta} = 0.89$, and TimesFM 2.5 the highest R^2 (0.235). Moirai-MoE-S undershoots more at this horizon ($\hat{\beta} = 0.48$), while Toto's severe undershoot ($\hat{\beta} = 0.26$) confirms its persistent scale misalignment. This confirms that the best TSFMs' advantage at longer horizons is not only in loss-function terms but also in the structural

quality of their forecasts.

Table 12: Forecast accuracy and Mincer–Zarnowitz efficiency ($h = 22$). Cross-sectional averages across 50 assets. The MZ regression is $RV_t = \alpha + \beta \widehat{RV}_t + \varepsilon_t$; under forecast efficiency, $\alpha = 0$ and $\beta = 1$. % Reject reports the fraction of assets for which the joint null is rejected at the 5% level (Wald test, Newey–West standard errors).

	Loss functions			MZ regression				
Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	$\hat{\alpha}$	$\hat{\beta}$	R^2	F-stat	% Reject
Econometric benchmarks								
HAR	0.310	1.46	4.865	0.0001	0.506	0.104	18.39	66%
HAR-J	0.316	1.45	4.776	0.0001	0.524	0.110	16.47	66%
HAR-RS	0.323	1.54	7.848	0.0002	0.387	0.091	31.62	86%
HARQ	0.316	1.45	11.441	0.0002	0.319	0.081	53.72	98%
Log-HAR	0.290	1.30	0.318	0.0001	0.503	0.082	16.05	86%
ARFIMA	0.466	1.35	0.271	0.0001	0.571	0.140	21.43	80%
Foundation models								
Chronos-Bolt-S	0.268	1.11	0.302	0.0001	1.121	0.202	10.99	96%
Chronos-Bolt-B	0.270	1.10	0.305	0.0000	1.453	0.221	12.16	98%
Moirai-2.0-S	0.264	1.06	0.250	0.0000	0.985	0.187	12.56	98%
Moirai-MoE-S	0.274	1.10	0.221	0.0001	0.483	0.203	78.71	98%
Lag-Llama	0.293	1.24	0.276	0.0001	0.705	0.056	3.42	50%
TimesFM-2.5	0.261	1.08	0.305	0.0000	1.301	0.235	12.83	98%
Toto	0.385	1.55	6.711	0.0002	0.255	0.172	741.81	98%
Sundial	0.234	1.06	0.204	0.0001	0.892	0.211	5.73	80%
TTM	0.249	1.26	0.191	0.0000	0.787	0.200	14.12	30%

Time-varying performance: GR fluctuation test. Figure 5 plots the rolling DM statistic (QLIKE loss) of each model against HAR at $h = 1$, averaged across 50 assets. A negative rolling DM indicates that the comparison model produces lower QLIKE than HAR during that window; values beyond the ± 2.80 critical-value bands indicate rejection of the equal predictive ability null at the 5% level.

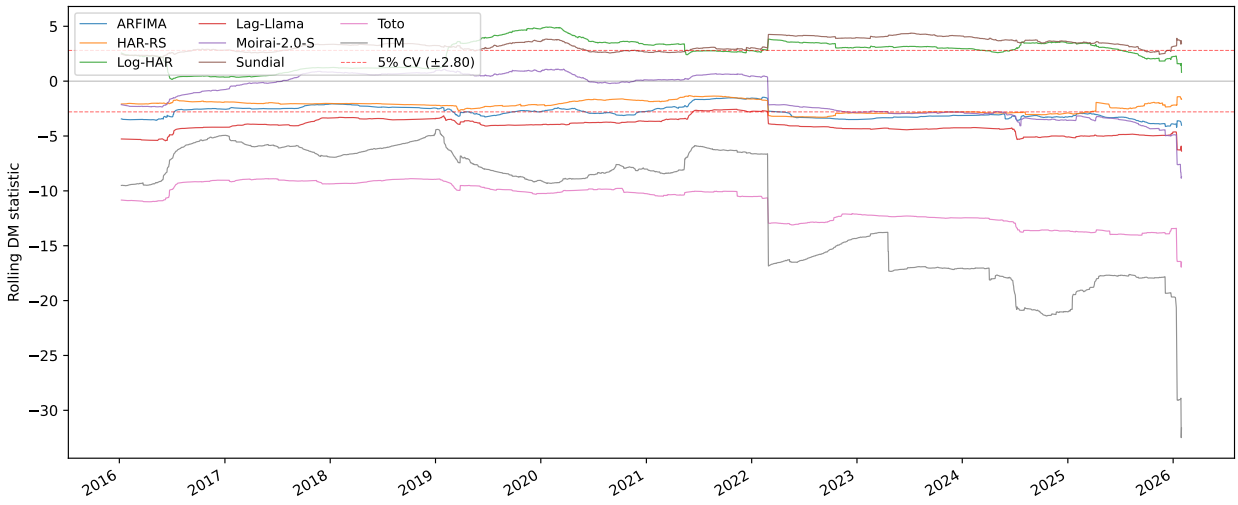

Figure 5: GR fluctuation test: rolling DM statistic (QLIKE loss) of each model vs. HAR at $h = 1$, averaged across 50 assets. Negative values indicate the comparison model outperforms HAR. Dashed red lines mark the 5% critical values (±2.80). The window size is 30% of the sample.

The GR tests reveal three patterns. First, the relative performance of all models against HAR is time-varying: the null of constant relative predictive ability is rejected for 100% of assets for Log-HAR, ARFIMA, Lag-Llama, Sundial, and Toto, and for 62–94% for the remaining TSFMs. This means that unconditional DM tests and MCS results, while informative about average performance, mask substantial temporal variation. Second, Log-HAR's advantage over HAR is concentrated in the 2019–2021 period, where it crosses above the upper critical value, indicating that it significantly *underperforms* HAR during the COVID-induced volatility spike. This is consistent with the log-space bias correction amplifying forecast errors during regime changes. After 2022, Log-HAR's advantage narrows and approaches zero. Third, TSFMs appear in the GR plot later than econometric models because the 512-day context requirement delays their first forecast to approximately 2017, and the GR rolling window (30% of the common sample) requires a further warm-up period before the first statistic can be computed. Once available, TSFMs immediately settle below the lower critical-value band, indicating persistent and significant outperformance of HAR.

Sundial shows the most stable advantage, with 100% rejection at $h = 1$.

Figures 6 and 7 present the corresponding plots at $h = 5$ and $h = 22$. At $h = 22$, the fluctuation test rejects less frequently (24–86% across models), consistent with the finding that forecast differences diminish at longer horizons as all models converge toward the unconditional mean (Patton, 2011).

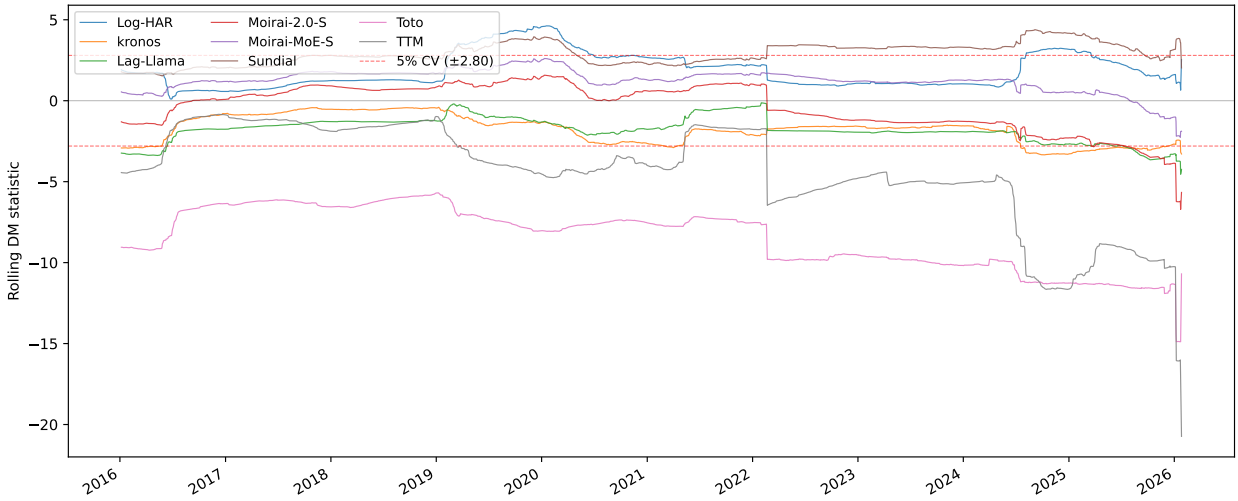

Figure 6: GR fluctuation test: rolling DM statistic (QLIKE loss) of each model vs. HAR at $h = 5$, averaged across 50 assets. Negative values indicate the comparison model outperforms HAR. Dashed red lines mark the 5% critical values (±2.80). The window size is 30% of the sample.

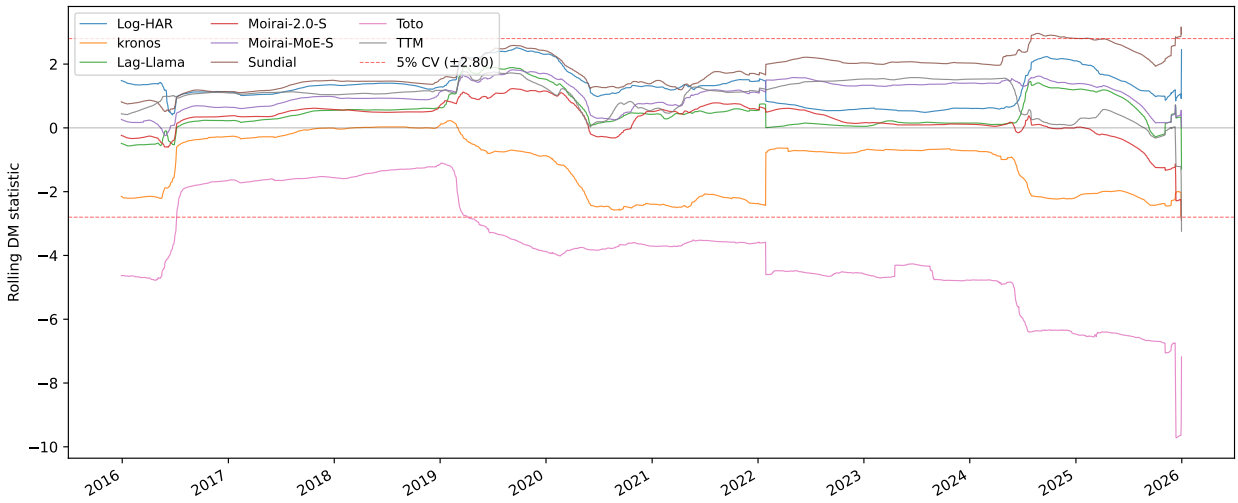

Figure 7: GR fluctuation test: rolling DM statistic (QLIKE loss) of each model vs. HAR at $h = 22$, averaged across 50 assets. Negative values indicate the comparison model outperforms HAR. Dashed red lines mark the 5% critical values (±2.80). The window size is 30% of the sample.

Interpretation. Taken together, the four tests support the same conclusion: the best TSFMs produce statistically superior and better-calibrated forecasts than HAR-family models, but the TSFM class is not uniformly strong. The MCS identifies Sundial as the model with the highest inclusion rate across all horizons (92–100%; Table 8), with Moirai-MoE-S as a consistent runner-up and TTM emerging at 86% at $h = 22$. The DM tests confirm that Sundial's win rate (88.7% at $h = 1$; Table 9) far exceeds all other models. The MZ regressions show that the best TSFMs maintain calibration at long horizons ($\hat{\beta}$ closer to 1; Tables 10 and 12), while econometric models suffer severe undershoot at $h = 22$. The GR test reveals that relative performance is not constant (Figure 5): Log-HAR's advantage over HAR erodes over time, and TSFMs consistently outperform HAR from their entry into the sample onward. The wide heterogeneity across TSFM architectures, from Sundial's consistent top ranking to Toto's miscalibration and TTM's horizon-dependent strength, suggests that pretraining data composition and output mechanism (flow matching vs. quantile

regression vs. MLP mixing) matter greatly.

7 Robustness Checks

The results in Sections 5 and 6 establish model rankings and their statistical significance. A natural question is whether these rankings are stable across market regimes, estimation choices, and evaluation parameters. We assess the stability of our main findings along several dimensions.

Sub-sample stability: pre- and post-COVID. Table 13 splits the full sample at March 1, 2020, and reports forecast accuracy separately for the pre-COVID (2015–2020) and post-COVID (2020–2026) periods across all 50 assets. The sub-sample analysis reveals that TSFM performance is stable across regimes, while levels-based econometric models are not.

In the pre-COVID period, Sundial achieves the lowest QLIKE at $h = 1$ (0.192) and $h = 5$ (0.166), while TTM achieves the lowest QLIKE at $h = 22$ (0.269), outperforming Log-HAR (0.291) and Sundial (0.331). The pre-COVID period features lower average volatility and fewer extreme events, yet Sundial already achieves lower QLIKE than all econometric models at $h = 1$ and $h = 5$ (Table 13). TTM is poor at $h = 1$ (QLIKE = 0.413) and $h = 5$ (0.294) in the pre-COVID period but transitions to best-in-class at $h = 22$, consistent with its full-sample horizon-dependent pattern. Toto remains miscalibrated in both periods (QLIKE 31.1 pre-COVID, 31.5 post-COVID at $h = 1$).

In the post-COVID period, the top TSFMs achieve the lowest loss across all metrics and horizons. Sundial achieves the lowest QLIKE at $h = 1$ (0.190 vs. 0.216 for Log-HAR), $h = 5$ (0.148 vs. 0.233), and $h = 22$ (0.137 vs. 0.317). TTM achieves a QLIKE of 0.150 at $h = 22$ in the post-COVID period, close to Sundial's 0.137 and well ahead of Log-HAR (0.317), confirming its strong monthly-horizon performance in both regimes. TimesFM 2.5 achieves the lowest MSE at $h = 1$, while Sundial leads on MSE at $h = 5$ and $h = 22$. The levels-based HAR variants suffer extreme QLIKE inflation in the high-volatility regime:

HARQ's QLIKE rises from 2.31 pre-COVID to 23.34 post-COVID at $h = 1$, because the near-zero forecast problem is exacerbated when realized values are large. Log-HAR is more stable across regimes than its levels-based counterparts but still falls behind the TSFMs on all metrics in the post-COVID sample. The TSFM heterogeneity pattern persists across sub-samples: Lag-Llama underperforms in both periods at $h = 1$ (QLIKE 0.417 pre-COVID, 0.553 post-COVID) but improves at longer horizons, and TTM is poor at $h = 1$ in both periods (0.413, 0.352) but competitive at $h = 22$. These results (Table 13) suggest that TSFMs' advantage is present in both calm and volatile market conditions, though the gap widens in the post-COVID regime where their flexible nonparametric forecasts adapt more effectively than the HAR's linear structure.

Table 13: Sub-sample stability: pre-COVID (before March 2020) and post-COVID (March 2020 onward) forecast accuracy across 50 assets (40 equities, 5 FX, 5 futures). Bold indicates best in each column within each panel. † denotes QLIKE values inflated by near-zero forecasts.

	Pre-COVID			Post-COVID		
Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE
Panel A: $h = 1$						
HAR	0.110	0.97	0.450	3.829	1.83	1.489†
HAR-J	0.098	0.94	0.407	3.859	1.81	1.314†
HAR-RS	0.282	1.09	1.223†	4.012	1.93	5.184†
HARQ	0.113	0.95	2.310†	3.975	1.89	23.341†
Log-HAR	0.089	0.82	0.199	3.682	1.59	0.216
ARFIMA	0.117	1.06	0.319	4.277	2.05	0.325
Chronos-Bolt-S	0.086	0.78	0.225	3.680	1.43	0.222
Chronos-Bolt-B	0.084	**0.77**	0.226	3.657	1.42	0.223
Moirai-2.0-S	0.085	0.81	0.258	3.716	1.44	0.239
Moirai-MoE-S	0.086	0.81	0.215	5.481	1.53	0.206
Lag-Llama	0.107	1.13	0.417	3.884	1.98	0.553
TimesFM-2.5	0.084	**0.77**	0.220	**3.611**	**1.40**	0.213
Toto	0.133	1.91	31.097†	3.938	2.70	31.520†
Sundial	**0.082**	0.80	**0.192**	3.654	1.48	**0.190**
TTM	0.091	1.46	0.413	3.707	2.21	0.352
Panel B: $h = 5$						
HAR	0.065	0.94	0.368	1.276	1.78	3.197†
HAR-J	0.054	0.91	0.429	1.353	1.75	2.970†
HAR-RS	0.204	1.05	1.849†	1.303	1.88	8.226†
HARQ	0.061	0.92	1.774†	1.364	1.80	23.476†
Log-HAR	0.041	0.78	0.170	1.248	1.54	0.233
ARFIMA	0.057	0.93	0.252	1.698	1.84	0.251
Chronos-Bolt-S	0.044	0.73	0.221	1.181	1.33	0.196
Chronos-Bolt-B	0.044	0.72	0.222	1.191	1.32	0.200
Moirai-2.0-S	0.043	0.74	0.214	1.217	1.30	0.193
Moirai-MoE-S	0.042	0.72	0.181	1.212	1.31	0.156
Lag-Llama	0.055	0.98	0.262	1.361	1.73	0.324
TimesFM-2.5	0.043	**0.72**	0.218	1.154	**1.28**	0.187
Toto	0.060	1.27	18.367†	1.186	1.90	13.714†
Sundial	**0.040**	**0.72**	**0.166**	**1.084**	1.32	**0.148**
TTM	0.045	1.20	0.294	1.149	1.90	0.252
Panel C: $h = 22$						
HAR	0.103	1.10	1.374†	0.611	1.79	6.571†
HAR-J	0.096	1.08	1.318†	0.623	1.77	6.486†
HAR-RS	0.133	1.16	3.275†	0.628	1.90	10.074†
HARQ	0.101	1.08	2.188†	0.617	1.77	17.475†
Log-HAR	**0.081**	0.99	0.291	0.569	1.53	0.317
ARFIMA	0.087	1.05	0.338	0.988	1.62	0.220
Chronos-Bolt-S	0.100	0.99	0.471	0.518	1.26	0.216
Chronos-Bolt-B	0.100	0.98	0.469	0.522	1.26	0.222
Moirai-2.0-S	0.098	0.96	0.385	0.512	**1.20**	0.184
Moirai-MoE-S	0.096	**0.94**	0.370	0.507	1.27	0.142
Lag-Llama	0.101	1.10	0.332	0.572	1.40	0.255
TimesFM-2.5	0.100	0.99	0.488	0.502	1.22	0.210
Toto	0.119	1.33	10.418†	0.659	1.74	4.754†
Sundial	0.093	**0.94**	0.331	**0.445**	**1.20**	**0.137**
TTM	0.091	1.08	**0.269**	0.483	1.44	0.150

Estimation window: 252 vs. 512 days. A potential concern is that TSFMs receive 512 days of context while econometric models use 252-day estimation windows, creating an information asymmetry that may favor TSFMs. To address this, we re-estimate all six

econometric benchmarks with a 512-day rolling window, matching the TSFM context length. Table 14 reports the results at $h = 1$ for the 50-asset sample. Five of six models improve under the longer window: HAR QLIKE drops from 1.07 to 0.21, and levels-HAR QLIKE inflation decreases substantially (e.g., HAR-RS QLIKE drops from 3.37 to 1.02). The exception is ARFIMA, whose long-memory estimation degrades with a 512-day window (QLIKE rises from 0.32 to 0.44). Cross-model QLIKE rankings are largely preserved (Spearman $\rho = 0.77$). Even after granting econometric models the same 512-day information set, the best TSFM zero-shot QLIKE values (Sundial at 0.190, Moirai-MoE-S at 0.209; Table 16) remain comparable to the best 512-day econometric QLIKE (Log-HAR at 0.191), confirming that the TSFM advantage is not an artifact of the longer context window. The same pattern holds at longer horizons: Tables 22 and 23 in Appendix A confirm that the 512-day window improves all HAR variants at $h = 5$ and $h = 22$, while ARFIMA deteriorates further. Notably, the 512-day Log-HAR QLIKE at $h = 5$ (0.155) essentially matches Sundial's cross-asset QLIKE (0.153), and at $h = 22$ the 512-day Log-HAR QLIKE (0.188) falls between TTM (0.191) and Sundial (0.204), indicating that much of the TSFM advantage at longer horizons can be recovered by granting econometric models the same information set.

Table 14: Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 1$, 50 assets). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^{-8}$)		MAE ($\times 10^{-4}$)		QLIKE	
Model	252	512	252	512	252	512
HAR	**159.09**	171.18	1.42	**1.36**	1.0679	**0.2090**
HAR-J	**159.44**	176.72	1.39	**1.33**	0.9377	**0.3774**
HAR-RS	176.91	**174.21**	1.54	**1.40**	3.3726	**1.0235**
HARQ	**166.09**	170.29	1.45	**1.37**	13.8459	**5.3843**
Log-HAR	**152.35**	159.24	1.23	**1.22**	0.2085	**0.1907**
ARFIMA	**176.61**	180.18	**1.57**	1.70	**0.3192**	0.4425

Context window length. Table 15 examines how TSFM forecast accuracy varies with context length (hereafter, ctx) of 128, 256, and 512 days for all nine TSFMs. Toto remains

severely miscalibrated at all context lengths (QLIKE > 31 at $h = 1$). TTM shows the strongest context dependence of any model: at ctx=128, QLIKE exceeds 1.4 at all horizons, indicating that 128 days of context is catastrophically insufficient for the MLP-Mixer architecture. Performance improves at ctx=256 (QLIKE $= 0.623$ at $h = 1$) and further at ctx=512 (QLIKE $= 0.373$), though only the 512-day context yields competitive QLIKE values. This monotonic improvement with context length is more pronounced than for any other TSFM and suggests that TTM's frequency prefix tuning requires a long input window to calibrate its internal representation of daily-frequency data. Among the remaining seven models, three patterns emerge. First, longer context generally improves Chronos-Bolt-S: QLIKE decreases from 0.231 (ctx=128) to 0.222 (ctx=512) at $h = 1$, with similar patterns at $h = 5$. Second, Moirai 2.0 shows a more nuanced pattern: while longer context improves QLIKE at $h = 1$ (0.264 to 0.243), it *worsens* performance at $h = 22$ (0.239 at ctx=128 vs. 0.251 at ctx=512), suggesting that Moirai's architecture captures long-range dependencies efficiently even from shorter windows and that additional context introduces noise at the monthly horizon. Third, Lag-Llama benefits substantially from longer context at $h = 5$ and $h = 22$ (QLIKE drops from 0.515 to 0.297 at $h = 5$), but at $h = 1$ the relationship is non-monotonic, with the best QLIKE at ctx=256 (0.379).

Sundial shows the most consequential context dependence: QLIKE improves monotonically with context length at all horizons, from 0.206 (ctx=128) to 0.190 (ctx=512) at $h = 1$ and from 0.239 to 0.204 at $h = 22$. Sundial's QLIKE at ctx=128 (0.206) is substantially worse than at ctx=512 (0.190), indicating that shorter context degrades its flow-matching architecture. TimesFM 2.5 is the least context-sensitive model: QLIKE varies by less than 0.01 across context lengths at $h = 1$ (0.222 at ctx=128 vs. 0.215 at ctx=512), with a similar pattern at $h = 22$ where the shorter contexts slightly outperform (0.302 at ctx=128 vs. 0.307 at ctx=512). Moirai-MoE-S shows moderate context sensitivity on QLIKE, with the best performance at ctx=256 for $h = 5$ and $h = 22$ (0.164 and 0.212, respectively), though the differences from ctx=512 are small. These results confirm that the 512-day default is

appropriate for most models but not universally optimal: Moirai 2.0 and Moirai-MoE-S prefer shorter context at the monthly horizon, TimesFM 2.5 is largely insensitive to context length, and both Sundial and TTM require the full 512-day window to produce competitive forecasts.

Table 15: Context-length sensitivity of TSFM forecasts across 50 assets (40 equities, 5 FX, 5 futures). QLIKE averaged across all assets, by horizon and context window length. Bold indicates the best context length for each model–horizon pair. Toto is omitted due to degenerate QLIKE values (> 5).

Model	h	ctx=128	ctx=256	ctx=512
Chronos-Bolt-S	1	0.231	0.232	**0.222**
	5	0.216	0.208	**0.203**
	22	**0.299**	0.310	0.303
Chronos-Bolt-B	1	0.401	0.244	**0.224**
	5	0.346	**0.206**	0.206
	22	0.394	0.307	**0.307**
Moirai-2.0-S	1	0.264	0.249	**0.243**
	5	0.201	0.199	**0.197**
	22	**0.239**	0.243	0.251
Moirai-MoE-S	1	0.221	0.212	**0.209**
	5	0.176	0.164	**0.164**
	22	0.219	**0.212**	0.222
Lag-Llama	1	0.433	**0.379**	0.464
	5	0.515	0.314	**0.297**
	22	0.587	0.338	**0.276**
TimesFM-2.5	1	0.222	0.215	**0.215**
	5	0.209	0.199	**0.196**
	22	**0.302**	0.302	0.307
Sundial	1	0.206	0.232	**0.190**
	5	0.180	0.183	**0.153**
	22	0.239	0.235	**0.204**
TTM	1	1.425	0.623	**0.373**
	5	1.807	0.935	**0.266**
	22	2.215	1.944	**0.191**

QLIKE floor sensitivity. Our QLIKE implementation floors forecasts at 10^{-10} to prevent division by zero. Because levels-based HAR variants occasionally produce near-zero forecasts,

this floor choice could affect model rankings. Table 16 reports cross-asset mean QLIKE at $h = 1$ under four floor values (10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}). QLIKE values are identical for floors $\leq 10^{-6}$ (Spearman rank correlation $\rho = 1.0$ vs. 10^{-10}), confirming that the floor never binds at these levels. Only the aggressive 10^{-4} floor alters rankings ($\rho = 0.87$) by clipping levels-HAR forecasts that drive QLIKE inflation (e.g., HARQ drops from 15.13 to 0.44). Our default 10^{-10} floor is conservative and does not distort model comparisons.

Table 16: QLIKE floor sensitivity across 50 assets (40 equities, 5 FX, 5 futures): cross-asset mean QLIKE at $h = 1$ under four floor values. Rankings are stable across floors (Spearman $\rho > 0.99$).

Model	$1e-10$	$1e-08$	$1e-06$	$1e-04$
HAR	1.14	1.14	1.14	0.3183
HAR-J	0.9797	0.9797	0.9797	0.3093
HAR-RS	3.51	3.51	3.51	0.3435
HARQ	15.13	15.13	15.13	0.4429
Log-HAR	0.2106	0.2106	0.2106	0.2992
ARFIMA	0.3162	0.3162	0.3162	0.3751
Chronos-Bolt-S	0.2221	0.2221	0.2221	0.2997
Chronos-Bolt-B	0.2239	0.2239	0.2239	0.2994
Moirai-2.0-S	0.2429	0.2429	0.2429	0.2948
Moirai-MoE-S	0.2094	0.2094	0.2094	0.2866
Lag-Llama	0.4637	0.4637	0.4637	0.4169
TimesFM-2.5	0.2151	0.2151	0.2151	0.2937
Toto	31.42	31.42	31.42	0.5219
Sundial	0.1903	0.1903	0.1903	0.2795
TTM	0.3728	0.3728	0.3728	0.3740

MZ bias-corrected evaluation. A model's low loss could reflect good calibration (i.e., alignment between the scale of forecasts and realizations) rather than genuine predictive information about future volatility dynamics. The MZ regression provides a natural correction: by regressing realized values on forecasts, one obtains intercept and slope parameters that remove any linear bias. If a model's accuracy survives this correction, its forecasts contain information beyond what a simple affine rescaling could produce. We apply MZ correction only to TSFM forecasts because the econometric models are estimated by OLS, which implic-

itly minimizes squared residuals and thus already produces approximately unbiased forecasts in-sample; applying MZ correction to OLS-estimated models would yield negligible changes. Specifically, we apply a recursive MZ bias correction to all TSFM forecasts at $h = 1$: at each time t, we estimate $\hat{\alpha}_t$ and $\hat{\beta}_t$ from all observations up to $t-1$, then form the corrected forecast $\widehat{RV}_t^{\text{MZ}} = \hat{\alpha}_t + \hat{\beta}_t \widehat{RV}_t$ using an expanding estimation window (minimum 252 days). Table 17 compares original and MZ-corrected TSFM forecasts against econometric benchmarks, evaluated on the common post-warm-up sample. MZ correction improves QLIKE for most well-calibrated models (TimesFM 2.5: 0.2160 $\rightarrow$ 0.1900; Chronos-Bolt-Base: 0.2242 $\rightarrow$ 0.1895), while Sundial's already-optimal QLIKE is essentially unchanged (0.1908 $\rightarrow$ 0.1917). MZ correction dramatically improves Toto (32.33 $\rightarrow$ 12.83), indicating that much of Toto's poor performance stems from severe scale misalignment rather than absence of signal. Lag-Llama worsens under correction, where the linear adjustment pushes some forecasts toward zero. Moirai-MoE-S improves modestly (0.2087 $\rightarrow$ 0.2056), while Moirai 2.0 is nearly unchanged (0.2431 $\rightarrow$ 0.2392), indicating that both Moirai variants are already well calibrated at this horizon. MSE is largely unchanged for the well-calibrated models, confirming that TSFM accuracy reflects genuine predictive content rather than calibration artifacts. We restrict the bias correction to $h = 1$ because at longer horizons the number of non-overlapping forecast observations shrinks (by factors of 5 and 22), making the recursive MZ estimates noisy during the early warm-up period and the correction less reliable.

Table 17: MZ bias-corrected TSFM evaluation across 50 assets (40 equities, 5 FX, 5 futures). Cross-asset mean metrics at $h = 1$.

Model	Variant	MSE ($\times 10^{-8}$)	QLIKE
HAR	Original	180.56	1.26
HAR-J	Original	181.82	1.07
HAR-RS	Original	189.81	3.87
HARQ	Original	188.92	16.84
Log-HAR	Original	174.39	0.2138
ARFIMA	Original	202.93	0.3227
Chronos-Bolt-S	Original	172.96	0.2234
Chronos-Bolt-S	MZ-Corrected	192.19	0.1945
Chronos-Bolt-B	Original	171.76	0.2242
Chronos-Bolt-B	MZ-Corrected	195.44	0.1895
Moirai-2.0-S	Original	174.30	0.2431
Moirai-2.0-S	MZ-Corrected	181.20	0.2392
Moirai-MoE-S	Original	252.52	0.2087
Moirai-MoE-S	MZ-Corrected	675.47	0.2056
Lag-Llama	Original	185.23	0.4807
Lag-Llama	MZ-Corrected	187.85	24.81
TimesFM-2.5	Original	169.53	0.2160
TimesFM-2.5	MZ-Corrected	204.64	0.1900
Toto	Original	188.40	32.33
Toto	MZ-Corrected	193.05	12.83
Sundial	Original	171.45	0.1908
Sundial	MZ-Corrected	208.31	0.1917
TTM	Original	174.20	0.3558
TTM	MZ-Corrected	244.40	0.5789

8 Conclusion

This paper evaluates nine zero-shot time series foundation models for realized volatility forecasting across 50 assets spanning U.S. equities, foreign exchange, and futures.

Three main findings emerge. First, the best zero-shot TSFMs outperform the best econometric benchmarks across all horizons and asset classes, contrasting with prior negative findings on daily returns and on earlier TSFM generations. Sundial achieves the lowest QLIKE at nearly every horizon and enters the MCS for the vast majority of assets, while TTM, the smallest model in the evaluation, achieves the lowest equity QLIKE at the monthly horizon

and enters the MCS for most assets at that horizon. Performance is highly heterogeneous across architectures, with some models producing forecasts with severe scale misalignment. Second, the TSFM advantage grows with forecast horizon: Sundial's QLIKE lead over Log-HAR widens substantially from the daily to the monthly horizon, and TTM's improvement from weak daily performance to best-in-class at the monthly horizon is the most dramatic horizon effect in our evaluation. Third, model selection within the TSFM class matters as much as the choice between TSFMs and econometric models.

Several limitations apply. We evaluate only zero-shot performance; fine-tuning on realized volatility data may yield further gains. Our evaluation is restricted to point forecasts (median of each model's predictive distribution); density evaluation using proper scoring rules (e.g., continuous ranked probability score) remains an open question, given that several TSFMs produce full predictive distributions. While no TSFM was trained on realized volatility, indirect exposure to related financial series cannot be fully excluded; temporal holdout designs could further address this concern. Extension to realized covariance forecasting and portfolio construction is a natural next step.

For practitioners, Sundial is the recommended default at daily and weekly horizons, achieving the lowest QLIKE across all asset classes with near-optimal calibration. At the monthly horizon, TTM is a strong alternative for equity volatility forecasting, achieving the lowest QLIKE and MSE among all models despite being the smallest and running efficiently on CPU. The TSFM advantage is most pronounced at the monthly horizon, where econometric models show substantially higher loss. For underperforming TSFMs, MZ bias correction can recover underlying signal. The wide performance dispersion across architectures cautions against treating TSFMs as a uniform class; evaluating multiple architectures before deployment remains essential.

References

Coen Adler, Yuxin Chang, Felix Draxler, Samar Abdi, and Padhraic Smyth. Beyond accuracy: Are time series foundation models well-calibrated? *arXiv preprint arXiv:2510.16060*, 2025.

Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

Torben G. Andersen and Tim Bollerslev. Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905, 1998. doi: 10.2307/2527343.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96 (453):42–55, 2001. doi: 10.1198/016214501750332965.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003. doi: 10.1111/1468-0262.00418.

Torben G. Andersen, Tim Bollerslev, and Francis X. Diebold. Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *Review of Economics and Statistics*, 89(4):701–720, 2007. doi: 10.1162/rest.89.4.701.

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. arXiv:2403.07815.

Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.

Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(2):253–280, 2002. doi: 10.1111/1467-9868.00336.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004. doi: 10.1093/jjfinec/nbh001.

Ole E. Barndorff-Nielsen, Peter Reinhard Hansen, Asger Lunde, and Neil Shephard. Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6):1481–1536, 2008. doi: 10.3982/ECTA6495.

Ole E. Barndorff-Nielsen, Silja Kinnebrock, and Neil Shephard. Measuring downside risk: Realised semivariance. In Tim Bollerslev, Jeffrey Russell, and Mark Watson, editors, *Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle*, pages 117–136. Oxford University Press, 2010.

Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016. doi: 10.1016/j.jeconom.2015.10.007.

Rafael Branco, Alexandre Rubesam, and Mauricio Zevallos. Forecasting realized volatility: Does anything beat linear models? *Journal of Empirical Finance*, 78:101524, 2024. doi: 10.1016/j.jempfin.2024.101524.

Alessio Brini and Giacomo Toscano. SpotV2Net: Multivariate intraday spot volatility forecasting via vol-of-vol-informed graph attention networks. *International Journal of Forecasting*, 41(3):1093–1111, 2025. doi: 10.1016/j.ijforecast.2024.11.004.

Andrea Bucci. Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531, 2020. doi: 10.1093/jjfinec/nbaa008.

Andrea Carriero, Davide Pettenuzzo, and Shubhranshu Shekhar. Macroeconomic forecasting with large language models. *arXiv preprint arXiv:2407.00890*, 2024.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023. doi: 10.1093/jjfinec/nbac020.

Fabrizio Cipollini, Giulia Cruciani, Giampiero M. Gallo, Alessandra Insana, Edoardo Otranto, and Fabio Spagnolo. VOLARE: VOLatility Archive for Realized Estimates. *arXiv preprint arXiv:2602.19732*, 2026.

Adam Clements and Daniel P. A. Preve. A practical guide to harnessing the HAR volatility model. *Journal of Banking & Finance*, 133:106285, 2021. doi: 10.1016/j.jbankfin.2021.106285.

Ben Cohen, Emaad Khwaja, Kan Wang, Charles Masson, Elise Ramé, Youssef Doubli, and Othmane Abou-Amal. Toto: Time series optimized transformer for observability. *arXiv preprint arXiv:2407.07874*, 2024.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jjfinec/nbp001.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference*

on Machine Learning (ICML), volume 235 of *Proceedings of Machine Learning Research*, pages 10148–10167. PMLR, 2024. arXiv:2310.10688.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

Vijay Ekambaram, Arindam Jati, Pankaj Dayama, Sumanta Mukherjee, Nam H. Nguyen, Wesley M. Gifford, Chandra Reddy, and Jayant Kalagnanam. Tiny time mixers (TTMs): Fast pre-trained models for enhanced zero/few-shot forecasting of multivariate time series. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2401.03955.

Azul Garza, Cristian Challu, and Max Mergenthaler-Canseco. TimeGPT-1. *arXiv preprint arXiv:2310.03589*, 2024.

John Geweke and Susan Porter-Hudak. The estimation and application of long memory time series models. *Journal of Time Series Analysis*, 4(4):221–238, 1983. doi: 10.1111/j.1467-9892.1983.tb00371.x.

Raffaella Giacomini and Barbara Rossi. Forecast comparisons in unstable environments. *Journal of Applied Econometrics*, 25(4):595–620, 2010. doi: 10.1002/jae.1177.

Anubha Goel, Puneet Pasricha, and Juho Kanniainen. Time-series foundation AI model for value-at-risk forecasting. *arXiv preprint arXiv:2410.11773*, 2024.

Anubha Goel, Puneet Pasricha, Martin Magris, and Juho Kanniainen. Foundation time-series AI model for realized volatility forecasting. *arXiv preprint arXiv:2505.11163*, 2025.

Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024. arXiv:2402.03885.

C. W. J. Granger and Roselyne Joyeux. An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1(1):15–29, 1980. doi: 10.1111/j.1467-9892.1980.tb00297.x.

Elias Søvik Gunnarsson, Håkon Ramon Isern, Aristidis Kaloudis, Morten Risstad, Benjamin Vigdel, and Sjur Westgaard. Prediction of realized volatility and implied volatility indices using AI and machine learning: A review. *International Review of Financial Analysis*, 93: 103221, 2024. doi: 10.1016/j.irfa.2024.103221.

Peter Reinhard Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011. doi: 10.3982/ECTA5771.

Peter Reinhard Hansen, Zhuo Huang, and Howard Howan Shek. Realized GARCH: A joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27 (6):877–906, 2012. doi: 10.1002/jae.1234.

Jonathan R. M. Hosking. Fractional differencing. *Biometrika*, 68(1):165–176, 1981. doi: 10.1093/biomet/68.1.165.

Stanisław Łaniewski and Robert Ślepaczuk. Evaluating the Chronos foundation model for daily stock index forecasting. In *Proceedings of the International Conference on Information Systems Development (ISD 2025)*, 2025. doi: 10.62036/ISD.2025.48.

Radmir Mishelevich Leushuis and Nicolai Petkov. Advances in forecasting realized volatility: A review of methodologies. *Financial Innovation*, 12:5, 2026. doi: 10.1186/s40854-025-00809-5.

Zhe Li, Xiangfei Qiu, Peng Chen, Yihang Wang, Hanyin Cheng, Yang Shu, Jilin Hu, Chenjuan Guo, Aoying Zhou, Christian S. Jensen, and Bin Yang. TSFM-Bench: A comprehensive and unified benchmark of foundation models for time series forecasting. *arXiv preprint arXiv:2410.11802*, 2024. KDD 2025.

Yuxuan Liang, Haomin Wen, Yuqi Nie, Yushan Jiang, Ming Jin, Dongjin Song, Shirui Pan, and Qingsong Wen. Foundation models for time series analysis: A tutorial and survey. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024. arXiv:2403.14735.

Chenghao Liu, Taha Aksu, Juncheng Liu, Xu Liu, Hanshu Yan, Quang Pham, Silvio Savarese, Doyen Sahoo, Caiming Xiong, and Junnan Li. Moirai 2.0: When less is more for time series forecasting. *arXiv preprint arXiv:2511.11698*, 2025a.

Xu Liu, Juncheng Liu, Gerald Woo, Taha Aksu, Yuxuan Liang, Roger Zimmermann, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. Moirai-moe: Empowering time series foundation models with sparse mixture of experts. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024a. arXiv:2410.10469.

Yong Liu, Haoran Zhang, Chenyu Li, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Timer: Generative pre-trained transformers are large time series models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024b. arXiv:2402.02368.

Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, Caiyin Yang, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Sundial: A family of highly capable time series foundation models. *arXiv preprint arXiv:2502.00816*, 2025b. ICML 2025 Oral (Top 1%).

Chuong Luong and Nikolai Dokuchaev. Forecasting of realised volatility with the random forests algorithm. *Journal of Risk and Financial Management*, 11(4):61, 2018. doi: 10.3390/jrfm11040061.

Ben A. Marconi. Time series foundation models for multivariate financial time series forecasting. *arXiv preprint arXiv:2507.07296*, 2025.

John A. Miller, Mohammed Aldosari, Farah Saeed, Nasid Habib Barna, Subas Rana, I. Bu-

dak Arpinar, and Ninghao Liu. A survey of deep learning and foundation models for time series forecasting. *arXiv preprint arXiv:2401.13912*, 2024.

Jacob A. Mincer and Victor Zarnowitz. The evaluation of economic forecasts. In Jacob A. Mincer, editor, *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, pages 1–46. National Bureau of Economic Research, New York, 1969.

Fernando Moreno-Pino and Stefan Zohren. DeepVol: Volatility forecasting from high-frequency data with dilated causal convolutions. *Quantitative Finance*, 24(8):1105–1127, 2024. doi: 10.1080/14697688.2024.2387222.

Whitney K. Newey and Kenneth D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

Quoc Anh Nguyen, Ce Guo, and Wayne Luk. Repurposing language models for FX volatility forecasting: A data-efficient and context-aware approach. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, 2025. doi: 10.1145/3768292.3770386.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

Andrew J. Patton and Kevin Sheppard. Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3):683–697, 2015. doi: 10.1162/REST_a_00503.

Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

Eghbal Rahimikia, Hao Ni, and Weiguan Wang. Re(Visiting) time series foundation models in finance. *arXiv preprint arXiv:2511.18578*, 2025.

Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, Mohammad Javad Darvishi Bayazi, George Adamopoulos, Roland Riachi, Nadhir Hassen, Marin Biloš, Sahil Garg, Anderson Schneider, Nicolas Chapados, Alexandre Drouin, Valentina Zantedeschi, Yuriy Nevmyvaka, and Irina Rish. Lag-llama: Towards foundation models for probabilistic time series forecasting. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.08278.

Oleksandr Shchur, Abdul Fatir Ansari, Caner Turkmen, Lorenzo Stella, Nick Erickson, Pablo Guerron, Michael Bohlke-Schneider, and Yuyang Wang. fev-bench: A realistic benchmark for time series forecasting. *arXiv preprint arXiv:2509.26468*, 2025.

Yu Shi, Zongliang Fu, Shuo Chen, Bohan Zhao, Wei Xu, Changshui Zhang, and Jian Li. Kronos: A foundation model for the language of financial markets. *arXiv preprint arXiv:2508.02739*, 2025.

Sébastien Valeyre and Sofiane Aboura. LLMs for time series: An application for single stocks and statistical arbitrage. *arXiv preprint arXiv:2412.09394*, 2024.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, pages 5998–6008, 2017. arXiv:1706.03762.

Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 53140–53164. PMLR, 2024. arXiv:2402.02592. Oral presentation.

Jiexia Ye, Yongzi Yu, Weiqi Zhang, Le Wang, Jia Li, and Fugee Tsung. Empowering time series analysis with foundation models: A comprehensive survey. *arXiv preprint arXiv:2405.02358*, 2024.

Chao Zhang, Yihuang Zhang, Mihai Cucuringu, and Zhongmin Qian. Volatility forecasting with machine learning and intraday commonality. *Journal of Financial Econometrics*, 22 (2):492–530, 2024. doi: 10.1093/jjfinec/nbad005.

Chao Zhang, Xingyue Pu, Mihai Cucuringu, and Xiaowen Dong. Graph-based methods for forecasting realized covariances. *Journal of Financial Econometrics*, 23(2):nbae026, 2025. doi: 10.1093/jjfinec/nbae026.

Zhuohang Zhu, Haodong Chen, Qiang Qu, and Vera Chung. FinCast: A foundation model for financial time-series forecasting. *arXiv preprint arXiv:2508.19609*, 2025. CIKM 2025.

A Supplementary Tables

This appendix collects additional results referenced in the main text. Table 18 reports cross-sectional median metrics. Tables 19, 20, and 21 provide per-asset QLIKE breakdowns at $h = 1$, $h = 5$, and $h = 22$ across all 50 assets. Tables 22 and 23 extend the estimation window sensitivity analysis to $h = 5$ and $h = 22$.

Table 18: Forecast accuracy for 40 U.S. equities (VOLARE), cross-sectional medians. Median aggregation is robust to outlier assets with degenerate forecasts. Bold indicates the best value in each column within each panel.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE
Panel: $h = 1$ (1 day)			
HAR	0.116	1.13	0.219
HAR-J	0.113	1.14	0.445
HAR-RS	0.124	1.19	$1.05^{\dagger}$
HARQ	0.173	1.24	$10.30^{\dagger}$
Log-HAR	0.139	1.12	0.208
ARFIMA	0.148	1.37	0.299
Chronos-Bolt-S	0.096	0.97	0.212
Chronos-Bolt-B	0.094	0.95	0.213
Moirai-2.0-S	0.092	0.98	0.209
Moirai-MoE-S	0.098	1.01	0.195
Lag-Llama	0.156	1.38	0.384
TimesFM-2.5	**0.090**	**0.95**	0.203
Toto	0.175	2.19	$36.25^{\dagger}$
Sundial	0.094	1.00	**0.181**
TTM	0.097	1.65	0.310
Panel: $h = 5$ (5 days)			
HAR	0.096	1.12	0.269
HAR-J	0.094	1.10	0.381
HAR-RS	0.109	1.19	$3.39^{\dagger}$
HARQ	0.126	1.17	$11.81^{\dagger}$
Log-HAR	0.111	1.08	0.205
ARFIMA	0.099	1.19	0.234
Chronos-Bolt-S	0.077	0.90	0.196
Chronos-Bolt-B	0.080	0.91	0.196
Moirai-2.0-S	0.073	**0.86**	0.172
Moirai-MoE-S	**0.070**	0.90	0.149
Lag-Llama	0.128	1.20	0.295
TimesFM-2.5	0.073	0.88	0.189
Toto	0.130	1.48	$17.72^{\dagger}$
Sundial	0.078	0.91	**0.149**
TTM	0.075	1.36	0.227
Panel: $h = 22$ (22 days)			
HAR	0.078	1.20	$3.24^{\dagger}$
HAR-J	0.076	1.20	$3.33^{\dagger}$
HAR-RS	0.082	1.27	$5.23^{\dagger}$
HARQ	0.089	1.25	$8.65^{\dagger}$
Log-HAR	0.081	1.11	0.296
ARFIMA	0.073	1.10	0.274
Chronos-Bolt-S	0.071	0.94	0.319
Chronos-Bolt-B	0.072	0.97	0.326
Moirai-2.0-S	0.069	**0.89**	0.249
Moirai-MoE-S	0.110	0.99	0.232
Lag-Llama	0.079	1.03	0.267
TimesFM-2.5	0.071	0.93	0.320
Toto	0.254	1.51	$7.06^{\dagger}$
Sundial	0.070	0.93	0.217
TTM	**0.066**	1.06	**0.186**

Table 19: Per-asset QLIKE loss at $h = 1$ across 50 assets (40 equities, 5 FX, 5 futures). Bold indicates the lowest (best) QLIKE in each row. Chr-Bolt-S = Chronos-Bolt-Small, Chr-Bolt-B = Chronos-Bolt-Base, Moirai-S = Moirai 2.0-Small, TFM-2.5 = TimesFM 2.5, MoE-S = Moirai-MoE-Small. HAR-J, HAR-RS, and HARQ are omitted as their QLIKE values are dominated by near-zero forecast artifacts (see Section 5.1).

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	MoE-S	Lag-Llama	TFM-2.5	Toto	Sundial	TTM
Stocks												
AAPL	0.1997	**0.1932**	0.4700	0.2278	0.2328	0.2067	0.2209	0.4122	0.2225	38.0542	0.1937	0.3219
ADBE	0.1570	**0.1507**	0.4327	0.1760	0.1794	0.1621	0.1601	0.3468	0.1749	36.5618	0.1516	0.2425
AMD	0.1532	**0.1500**	0.2883	0.1697	0.1725	0.1633	0.1717	0.2888	0.1660	8.2771	0.1509	0.1751
AMGN	0.2145	0.2106	0.3082	0.2409	0.2351	0.2193	0.2100	0.3873	0.2311	37.4056	**0.2010**	0.3071
AMZN	13.8320	0.4528	0.6815	0.5050	0.5127	0.4790	0.4676	0.8656	0.4895	34.1204	**0.4157**	0.5222
AXP	0.2137	0.2241	0.2973	0.2111	0.2106	0.1959	0.1894	0.4140	0.2020	37.9825	**0.1762**	0.3119
BA	9.3426	0.2843	0.3670	0.2439	0.2447	0.2248	0.2326	0.4943	0.2406	32.5503	**0.2041**	0.2707
CAT	0.1769	0.1793	0.2530	0.1895	0.1907	0.1693	0.1735	0.3422	0.1850	38.2613	**0.1615**	0.2477
CRM	0.2299	0.2170	0.3759	0.2516	0.2520	0.2341	0.2232	0.3983	0.2413	34.5271	**0.2072**	0.2870
CSCO	0.1911	0.1852	0.3172	0.2102	0.2098	0.1913	0.1936	0.3785	0.2022	35.9292	**0.1798**	0.3258
CVX	0.1870	0.1962	0.2593	0.1583	0.1566	0.1658	0.1468	0.3065	0.1540	35.5364	**0.1414**	0.2748
DIS	2.5596	0.2216	0.3013	0.2302	0.2345	0.2058	0.2084	0.4009	0.2242	38.6116	**0.1937**	0.3207
GE	0.2007	0.1949	0.2884	0.2173	0.2185	0.2025	0.2008	0.3640	0.2104	29.8779	**0.1850**	0.2529
GOOGL	0.3662	0.3088	0.4195	0.3416	0.3519	0.3140	0.3223	0.5129	0.3408	42.1751	**0.2902**	0.3887
GS	0.1670	0.1643	0.2697	0.1739	0.1715	0.1529	0.1612	0.3189	0.1661	40.4164	**0.1505**	0.2402
HD	0.1971	0.1975	0.2923	0.1957	0.1939	0.1836	0.1837	0.4202	0.1847	41.1524	**0.1620**	0.3080
HON	0.2256	0.2227	0.3146	0.2078	0.2058	0.2115	0.1826	0.4032	0.1959	37.3624	**0.1738**	0.3454
IBM	0.6798	0.1981	0.3104	0.2133	0.2157	0.2109	0.1879	0.3832	0.2044	37.7081	**0.1823**	0.3357
JNJ	0.2310	0.2194	0.2925	0.2152	0.2128	0.2529	0.2006	0.3978	0.2095	33.4378	**0.1892**	0.3631
JPM	0.1874	0.1848	0.3035	0.1950	0.1943	0.1768	0.1755	0.3618	0.1883	35.8822	**0.1648**	0.3007
KO	0.1827	0.1767	0.2511	0.1675	0.1655	0.2524	0.1493	0.3446	0.1570	28.8196	**0.1435**	0.3737
MCD	3.9225	0.2386	0.2976	0.2252	0.2259	0.2622	0.2056	0.3831	0.2192	31.4435	**0.1945**	0.3867
META	0.2198	0.1855	0.3188	0.1887	0.1951	0.1790	0.1766	0.3721	0.1876	35.4874	**0.1649**	0.2564
MMM	0.2185	0.2121	0.2965	0.2271	0.2280	0.2509	0.2073	0.3729	0.2232	43.3445	**0.1916**	0.3196
MRK	0.2064	0.2000	0.2678	0.2212	0.2184	0.1922	0.1962	0.3507	0.2073	36.7627	**0.1809**	0.3035
MSFT	0.2980	0.2718	0.4233	0.3311	0.3256	0.3193	0.2891	0.4759	0.3122	41.9633	**0.2679**	0.3981
NFLX	10.5396	0.3476	0.4306	0.3809	0.3801	0.3696	0.3658	0.5480	0.3673	21.9215	**0.3224**	0.3875
NKE	0.2170	0.2098	0.2835	0.2105	0.2125	0.1818	0.1897	0.3396	0.2019	41.4459	**0.1805**	0.2800
NVDA	0.1909	0.1827	0.3225	0.2017	0.2062	0.1907	0.1927	0.3857	0.1995	19.9380	**0.1753**	0.2147
ORCL	1.6483	0.2585	0.3809	0.2854	0.2961	0.2608	0.2701	0.5327	0.2806	36.6540	**0.2368**	0.3503
PG	0.2001	0.1900	0.2404	0.1812	0.1806	0.2332	0.1630	0.3771	0.1740	28.2630	**0.1572**	0.3628
PM	0.2589	0.2424	0.2899	0.2549	0.2589	0.2433	0.2349	0.4061	0.2558	41.6249	**0.2280**	0.3335
SHW	1.5060	0.2060	0.2484	0.1991	0.2005	0.1784	0.1797	0.3264	0.1922	42.8356	**0.1719**	0.2768
TRV	0.1689	0.1655	0.2349	0.1726	0.1691	0.1858	0.1556	0.3279	0.1650	33.9876	**0.1492**	0.2832
TSLA	2.2254	0.1722	0.2929	0.2029	0.2036	0.1869	0.1967	0.3343	0.1974	13.8548	**0.1697**	0.1879
UNH	0.2226	0.2147	0.3134	0.2519	0.2527	0.2228	0.2269	0.4056	0.2401	38.4409	**0.2080**	0.2952
V	0.8008	0.2616	0.3140	0.2686	0.2732	0.3309	0.2393	0.4384	0.2644	33.9561	**0.2373**	0.3836
VZ	1.1632	0.1859	0.2658	0.1998	0.2035	0.2040	0.1888	0.3847	0.1945	34.9943	**0.1786**	0.3475
WMT	0.9400	0.2522	0.3115	0.2517	0.2498	0.2743	0.2280	0.4637	0.2481	35.1741	**0.2187**	0.3764
XOM	0.2045	0.2264	0.2449	0.1709	0.1678	0.2122	0.1559	0.3034	0.1615	36.7170	**0.1511**	0.2895
Average	1.3761	0.2189	0.3218	0.2292	0.2302	0.2263	0.2106	0.4018	0.2221	34.5865	**0.1951**	0.3137
FX												
AUDUSD	0.1617	0.1524	0.2132	0.1504	0.1536	0.3086	0.1487	0.2664	0.1482	16.6629	**0.1366**	0.5896
EURUSD	0.1571	0.1491	0.2035	0.1576	0.1594	0.3139	0.1988	0.2768	0.1560	11.1293	**0.1393**	0.8588
GBPUSD	0.1504	0.1376	0.2120	0.1492	0.1487	0.3198	0.1667	0.2806	0.1437	12.8796	**0.1327**	0.7736
USDCAD	0.1380	0.1308	0.1749	0.1326	0.1331	0.2587	0.1824	0.2376	0.1310	9.5052	**0.1229**	0.9215
USDJPY	0.2753	0.2435	0.4040	0.2823	0.2856	0.4456	0.2968	0.4621	0.2754	13.0623	**0.2350**	0.9012
Average	0.1765	0.1627	0.2415	0.1744	0.1761	0.3293	0.1987	0.3047	0.1709	12.6479	**0.1533**	0.8090
Futures												
C	0.3153	0.2903	0.3282	0.3256	0.3269	0.2969	0.3023	0.5447	0.3200	37.1299	**0.2765**	0.3751
CL	0.1998	0.1358	0.3310	0.1591	0.1521	0.1741	**0.1358**	3.0510	0.1363	24.1237	0.1483	0.3700
ES	0.2286	0.1976	0.5399	0.2244	0.2224	0.5307	0.2222	1.2473	0.2108	24.6635	**0.1927**	0.5771
GC	0.1648	**0.1601**	0.2879	0.1763	0.2223	0.2669	0.2117	0.3741	0.1734	28.8041	0.1612	0.5381
NG	0.2176	0.1787	0.2421	0.1790	0.1802	0.1773	0.1824	0.3715	0.1785	9.7896	**0.1654**	0.1922
Average	0.2252	0.1925	0.3458	0.2129	0.2208	0.2892	0.2109	1.1177	0.2038	24.9021	**0.1888**	0.4105
Overall Average	1.1411	0.2106	0.3162	0.2221	0.2239	0.2429	0.2094	0.4637	0.2151	31.4242	**0.1903**	0.3729

Table 20: Per-asset QLIKE loss at $h = 5$ across 50 assets (40 equities, 5 FX, 5 futures). Bold indicates the lowest (best) QLIKE in each row. Model abbreviations as in Table 19.

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	MoE-S	Lag-Llama	TFM-2.5	Toto	Sundial	TTM
Stocks												
AAPL	0.1783	**0.1668**	0.3870	0.2314	0.2385	0.1927	0.1757	0.2922	0.2280	20.1222	0.1728	0.2376
ADBE	0.1264	**0.1173**	0.3805	0.1643	0.1699	0.1406	0.1308	0.2420	0.1608	13.5529	0.1235	0.1745
AMD	0.1108	0.1086	0.2138	0.1389	0.1430	0.1336	0.1301	0.1725	0.1353	0.4828	**0.1073**	0.1231
AMGN	0.2248	0.2053	0.2356	0.2363	0.2345	0.2044	0.1818	0.3123	0.2290	18.5348	**0.1801**	0.2319
AMZN	19.2609	0.4705	0.5729	0.5106	0.5179	0.4695	0.4345	0.5694	0.4991	8.2384	**0.3686**	0.4147
AXP	0.2035	0.2807	0.2350	0.2013	0.1992	0.1616	0.1486	0.3605	0.1884	16.9035	**0.1456**	0.2278
BA	31.7150	0.4817	0.3190	0.2305	0.2268	0.1941	**0.1651**	0.4017	0.2226	15.5799	0.1695	0.1945
CAT	0.1398	0.1436	0.1792	0.1586	0.1626	0.1321	0.1268	0.2276	0.1566	15.7173	**0.1210**	0.1677
CRM	0.2246	0.2228	0.2950	0.2551	0.2554	0.2227	0.2011	0.2723	0.2501	15.0694	**0.1898**	0.2234
CSCO	0.9244	0.1683	0.2543	0.1986	0.1988	0.1596	**0.1477**	0.2837	0.1922	20.0926	0.1490	0.2333
CVX	2.5162	0.2777	0.2162	0.1504	0.1523	0.1339	0.1253	0.3232	0.1462	16.8956	**0.1246**	0.2028
DIS	8.5696	0.2826	0.2174	0.2182	0.2263	0.1704	**0.1616**	0.3200	0.2161	16.3909	0.1657	0.2272
GE	0.1648	0.1627	0.2036	0.1908	0.1885	0.1653	0.1502	0.2595	0.1862	12.3244	**0.1417**	0.1797
GOOGL	2.1229	0.2661	0.3177	0.3076	0.3215	0.2700	0.2476	0.3457	0.3073	17.5032	**0.2292**	0.2826
GS	0.1475	0.1570	0.2097	0.1515	0.1544	0.1238	**0.1190**	0.2465	0.1485	18.5384	0.1237	0.1665
HD	0.1990	0.2091	0.2376	0.1875	0.1869	0.1533	0.1399	0.3327	0.1758	18.9064	**0.1384**	0.2260
HON	0.2820	0.3500	0.2460	0.2014	0.1968	0.1730	**0.1479**	0.3479	0.1894	19.6790	0.1523	0.2499
IBM	1.6441	0.2038	0.2341	0.2044	0.2085	0.1654	**0.1471**	0.2969	0.1952	18.1382	0.1535	0.2344
JNJ	0.2565	0.2141	0.2343	0.1922	0.1909	0.1935	0.1517	0.2918	0.1856	20.3931	**0.1506**	0.2541
JPM	0.1760	0.1972	0.2489	0.1943	0.1959	0.1488	0.1434	0.3191	0.1853	19.5637	**0.1426**	0.2182
KO	0.1661	0.1815	0.2126	0.1666	0.1670	0.2116	**0.1201**	0.3091	0.1523	18.3551	0.1235	0.2648
MCD	5.7174	0.2857	0.2302	0.2107	0.2152	0.1992	0.1607	0.3788	0.2024	20.2296	**0.1584**	0.2818
META	3.0672	0.1969	0.2529	0.1829	0.1842	0.1618	0.1423	0.2561	0.1789	14.3516	**0.1399**	0.1907
MMM	0.6220	0.1950	0.2194	0.2026	0.2040	0.2049	0.1587	0.2632	0.2021	18.9966	**0.1513**	0.2192
MRK	0.1776	0.1687	0.1848	0.1917	0.1898	0.1464	0.1417	0.2338	0.1809	19.0289	**0.1372**	0.2001
MSFT	0.2953	0.2437	0.3282	0.2984	0.3080	0.2600	0.2397	0.3749	0.2926	17.6576	**0.2221**	0.2984
NFLX	14.9187	0.3343	0.3298	0.3395	0.3433	0.3230	0.3027	0.3636	0.3272	5.3863	**0.2632**	0.3015
NKE	0.2098	0.2113	0.2042	0.1934	0.1956	0.1476	**0.1442**	0.2729	0.1826	16.7470	0.1487	0.1991
NVDA	0.1693	0.1642	0.2522	0.1836	0.1878	0.1700	0.1614	0.2513	0.1880	3.8539	**0.1421**	0.1643
ORCL	4.6326	0.2788	0.3143	0.2984	0.3103	0.2443	0.2252	0.3914	0.2904	15.6476	**0.2224**	0.2597
PG	0.2826	0.2035	0.1889	0.1569	0.1606	0.1858	**0.1189**	0.2956	0.1510	19.5454	0.1227	0.2577
PM	0.2212	0.1903	0.2151	0.2176	0.2217	0.1803	**0.1715**	0.2951	0.2186	21.1392	0.1739	0.2284
SHW	1.6041	0.2297	0.1700	0.1677	0.1711	0.1413	**0.1280**	0.2595	0.1620	19.3818	0.1293	0.1934
TRV	0.1615	0.1726	0.1762	0.1513	0.1495	0.1431	**0.1171**	0.2865	0.1452	17.7860	0.1184	0.1935
TSLA	0.1585	0.1437	0.2219	0.1931	0.1950	0.1746	0.1720	0.2378	0.1907	1.6467	**0.1398**	0.1427
UNH	0.1972	0.1961	0.2286	0.2349	0.2334	0.1785	0.1676	0.2883	0.2203	14.9049	**0.1640**	0.2040
V	1.1644	0.2749	0.2398	0.2532	0.2574	0.2197	**0.1936**	0.3390	0.2466	18.2304	0.1970	0.2847
VZ	1.5620	0.1697	0.1933	0.1731	0.1814	0.1525	0.1400	0.2606	0.1705	21.6736	**0.1366**	0.2423
WMT	0.9095	0.2674	0.2456	0.2355	0.2353	0.2188	0.1905	0.2985	0.2345	21.7110	**0.1879**	0.2806
XOM	2.4919	0.3536	0.2074	0.1605	0.1594	0.1612	**0.1223**	0.3179	0.1481	16.6871	0.1254	0.2120
Average	2.6979	0.2287	0.2513	0.2134	0.2160	0.1883	0.1673	0.3048	0.2071	16.1397	**0.1613**	0.2272
FX												
AUDUSD	0.1179	0.1173	0.1485	0.1012	0.1039	0.2264	0.0934	0.1806	0.1007	13.8959	**0.0820**	0.4100
EURUSD	0.1039	0.1003	0.1416	0.1005	0.1024	0.2346	0.1333	0.1678	0.0999	10.0304	**0.0756**	0.6243
GBPUSD	0.1762	0.0998	0.1655	0.1167	0.1176	0.2573	0.1295	0.1877	0.1155	10.8819	**0.0953**	0.5690
USDCAD	0.0914	0.0879	0.1142	0.0828	0.0844	0.1897	0.1188	0.1487	0.0822	8.4382	**0.0669**	0.6833
USDJPY	0.2233	0.1953	0.3177	0.2375	0.2388	0.3421	0.2200	0.2867	0.2319	11.5840	**0.1714**	0.6540
Average	0.1425	0.1201	0.1775	0.1277	0.1294	0.2500	0.1390	0.1943	0.1260	10.9661	**0.0982**	0.5882
Futures												
C	0.1985	0.1772	0.2020	0.2159	0.2212	0.1851	0.1741	0.2632	0.2100	17.2483	**0.1573**	0.2361
CL	0.2014	0.1645	0.2832	0.1672	0.1746	0.1772	0.1214	0.4519	0.1430	7.5899	**0.1158**	0.2073
ES	1.4887	0.2531	0.4722	0.2805	0.2751	0.3454	0.2146	0.5195	0.2618	19.5806	**0.2077**	0.4525
GC	0.1216	0.1057	0.2129	0.1297	0.1727	0.1952	0.1392	0.1886	0.1279	18.8359	**0.1043**	0.2625
NG	0.1599	0.1231	0.1729	0.1322	0.1354	0.1301	0.1270	0.2409	0.1332	1.3642	**0.1112**	0.1369
Average	0.4340	0.1647	0.2687	0.1851	0.1958	0.2066	0.1553	0.3328	0.1752	12.9238	**0.1393**	0.2591
Overall Average	2.2160	0.2114	0.2457	0.2020	0.2053	0.1963	0.1633	0.2965	0.1958	15.3007	**0.1528**	0.2665

Table 21: Per-asset QLIKE loss at $h = 22$ across 50 assets (40 equities, 5 FX, 5 futures). Bold indicates the lowest (best) QLIKE in each row. Model abbreviations as in Table 19.

Asset	HAR	Log-HAR	ARFIMA	Chr-Bolt-S	Chr-Bolt-B	Moirai-S	MoE-S	Lag-Llama	TFM-2.5	Toto	Sundial	TTM
Stocks												
AAPL	0.1969	0.1808	0.3361	0.3136	0.3193	0.2447	0.2180	0.2470	0.3189	9.5453	0.1995	**0.1805**
ADBE	0.1398	**0.1348**	0.3791	0.2408	0.2470	0.2039	0.1721	0.1963	0.2474	7.3768	0.1601	0.1430
AMD	0.0897	0.0884	0.1856	0.1324	0.1356	0.1319	0.1203	0.1190	0.1314	0.1285	**0.0808**	0.0890
AMGN	3.2254	0.2634	0.2177	0.3563	0.3541	0.2988	0.2536	0.2610	0.3605	3.4688	0.2416	**0.1966**
AMZN	23.0166	0.4877	0.4683	0.5492	0.5540	0.4759	0.4012	0.4003	0.5499	2.7585	0.2977	**0.2668**
AXP	4.1452	0.3781	0.3191	0.3952	0.3945	0.2702	0.2561	0.3868	0.3731	7.7764	0.2497	**0.2193**
BA	48.4755	1.1024	0.4859	0.4640	0.4634	0.4147	0.3316	0.4528	0.4860	5.3179	0.3265	**0.2544**
CAT	4.6059	0.2131	0.1892	0.2444	0.2438	0.1883	0.1698	0.1931	0.2436	7.7874	0.1588	**0.1390**
CRM	11.4276	0.3410	0.2683	0.2891	0.2881	0.2422	0.2044	0.2096	0.2921	6.9779	0.1761	**0.1628**
CSCO	2.6118	0.2628	0.3505	0.2953	0.2982	0.2257	0.2149	0.2550	0.3016	6.5477	0.2048	**0.1881**
CVX	16.7375	0.7459	0.3206	0.3418	0.3423	0.2794	0.2661	0.3964	0.3527	11.6332	0.2629	**0.2244**
DIS	16.3347	0.5921	0.1851	0.3167	0.3324	0.2301	0.2111	0.3033	0.3306	7.4460	0.2175	**0.1818**
GE	0.1934	0.2004	0.2111	0.2373	0.2380	0.1991	0.1665	0.2203	0.2434	4.5332	0.1542	**0.1487**
GOOGL	6.7952	0.2986	0.2545	0.3646	0.3818	0.3120	0.2677	0.2549	0.3959	4.3320	0.2299	**0.1936**
GS	0.2048	0.2145	0.2868	0.2503	0.2558	0.2007	0.1890	0.2303	0.2576	6.0486	0.1974	**0.1548**
HD	1.4765	0.3321	0.3579	0.3579	0.3567	0.2690	0.2614	0.3500	0.3536	8.1227	0.2459	**0.2215**
HON	6.7514	0.5217	0.3450	0.3785	0.3761	0.2563	0.2470	0.3582	0.3561	9.7074	0.2575	**0.2164**
IBM	2.1795	0.2645	0.2330	0.3178	0.3269	0.2153	0.1963	0.2770	0.3070	6.6203	0.2144	**0.1748**
JNJ	0.3284	0.2763	0.2872	0.3486	0.3507	0.2890	0.2620	0.2817	0.3434	8.2663	0.2423	**0.2091**
JPM	6.3848	0.3178	0.3438	0.3719	0.3691	0.2749	0.2633	0.3335	0.3726	7.6711	0.2447	**0.2163**
KO	3.2457	0.3462	0.3343	0.3918	0.3967	0.2948	0.2696	0.3511	0.3699	9.4222	0.2776	**0.2203**
MCD	4.4248	0.4672	0.3247	0.4223	0.4230	0.3013	0.3124	0.4126	0.4320	9.6505	0.2971	**0.2557**
META	4.1675	0.2438	0.2527	0.2389	0.2411	0.2123	0.1746	0.2056	0.2614	5.3226	0.1611	**0.1509**
MMM	1.9908	0.2210	0.2090	0.2714	0.2690	0.2264	0.1825	0.2122	0.2807	6.7874	0.1814	**0.1538**
MRK	0.2130	0.1894	0.1718	0.2568	0.2582	0.1788	0.1750	0.2079	0.2568	5.9427	0.1692	**0.1411**
MSFT	4.6384	0.2857	0.3099	0.3927	0.4017	0.3111	0.2696	0.3071	0.4059	7.1386	0.2430	**0.2291**
NFLX	9.5100	0.3088	0.2297	0.3269	0.3278	0.3115	0.2783	0.2546	0.3337	0.8458	**0.2161**	0.2239
NKE	2.9300	0.3044	0.2201	0.2841	0.2867	0.2167	0.2054	0.2653	0.2804	7.7602	0.2044	**0.1760**
NVDA	0.2246	0.2052	0.2805	0.2322	0.2359	0.2086	0.1842	0.2019	0.2497	0.6064	0.1476	**0.1434**
ORCL	7.1831	0.3691	0.3904	0.4655	0.4867	0.3851	0.3491	0.3491	0.4817	7.3199	0.2941	**0.2243**
PG	3.1120	0.3395	0.3016	0.3307	0.3344	0.2721	0.2400	0.3286	0.3210	9.5718	0.2466	**0.2242**
PM	0.2639	0.2057	0.2625	0.2900	0.2941	0.2525	0.2346	0.2789	0.3214	5.5608	0.2172	**0.1827**
SHW	5.1294	0.2928	0.1803	0.2545	0.2588	0.2181	0.1946	0.2464	0.2674	7.5363	0.1836	**0.1710**
TRV	0.2825	0.3321	0.2575	0.2930	0.2850	0.2293	0.2295	0.2953	0.2972	5.0933	0.2172	**0.1781**
TSLA	0.1561	0.1660	0.1848	0.2469	0.2520	0.2213	0.2004	0.1921	0.2502	0.1654	0.1536	**0.1309**
UNH	3.1724	0.2263	0.2006	0.3292	0.3209	0.2221	0.1960	0.2567	0.3117	5.3770	0.2072	**0.1746**
V	3.8648	0.4143	0.2704	0.3721	0.3807	0.2710	0.2494	0.3090	0.3743	8.5910	0.2505	**0.2299**
VZ	1.9827	0.2200	0.2335	0.2702	0.2756	0.2068	0.2097	0.2390	0.2798	6.3886	0.1956	**0.1841**
WMT	6.0957	0.4762	0.2769	0.3359	0.3330	0.2550	0.2451	0.2697	0.3309	9.1863	0.2321	**0.2193**
XOM	19.0392	0.9555	0.3003	0.3194	0.3247	0.2565	0.2343	0.3941	0.3190	7.2077	0.2479	**0.2170**
Average	5.9237	0.3446	0.2804	0.3223	0.3253	0.2568	0.2327	0.2826	0.3261	6.4385	0.2176	**0.1903**
FX												
AUDUSD	0.1665	0.1624	0.1645	0.1441	0.1478	0.2003	0.1151	0.1739	0.1467	10.7784	**0.1081**	0.1486
EURUSD	0.1267	0.1306	0.1469	0.1232	0.1284	0.2007	0.1284	0.1409	0.1232	8.4332	**0.0850**	0.1836
GBPUSD	0.2949	0.1254	0.1719	0.1479	0.1485	0.2240	0.1515	0.1533	0.1514	9.7204	**0.1133**	0.1900
USDCAD	0.1115	0.1092	0.1240	0.1158	0.1196	0.1637	0.1123	0.1329	0.1172	7.3388	**0.0847**	0.1937
USDJPY	0.2295	0.2081	0.3196	0.2741	0.2714	0.2701	0.2157	0.2166	0.2852	9.8760	**0.1690**	0.2254
Average	0.1858	0.1471	0.1854	0.1610	0.1631	0.2117	0.1446	0.1635	0.1647	9.2293	**0.1120**	0.1882
Futures												
C	0.1695	0.1435	0.1502	0.2023	0.2040	0.1563	0.1342	0.1853	0.1933	7.1465	**0.1176**	0.1376
CL	0.3207	0.4053	0.4021	0.2964	0.2970	0.2697	0.1854	0.5906	0.2564	2.6843	**0.1854**	0.2381
ES	4.5906	0.6009	0.5021	0.5946	0.6009	0.4383	0.4187	0.5478	0.5986	12.4722	0.3753	**0.3526**
GC	0.1340	**0.1209**	0.2085	0.1683	0.1770	0.1760	0.1516	0.1553	0.1691	9.3199	0.1262	0.1525
NG	0.1661	0.1317	0.1653	0.1477	0.1520	0.1478	0.1377	0.2034	0.1584	0.2368	**0.1152**	0.1286
Average	1.0762	0.2805	0.2857	0.2819	0.2862	0.2376	0.2055	0.3365	0.2752	6.3719	**0.1839**	0.2019
Overall Average	4.8651	0.3185	0.2714	0.3021	0.3052	0.2504	0.2212	0.2761	0.3048	6.7109	0.2037	**0.1912**

Table 22: Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 5$, 50 assets). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^{-8}$)		MAE ($\times 10^{-4}$)		QLIKE	
Model	252	512	252	512	252	512
HAR	**56.16**	59.06	1.38	**1.26**	2.0277	**0.1744**
HAR-J	**58.35**	65.40	1.34	**1.24**	1.9189	**0.9094**
HAR-RS	66.00	**59.76**	1.50	**1.30**	5.5067	**0.6447**
HARQ	60.05	**58.23**	1.38	**1.27**	13.6673	**5.6516**
Log-HAR	53.77	**52.77**	1.18	**1.13**	0.2054	**0.1549**
ARFIMA	71.70	**68.95**	**1.39**	1.55	**0.2481**	0.3801

Table 23: Estimation window sensitivity: 252-day vs. 512-day rolling window ($h = 22$, 50 assets). Cross-asset mean metrics. Bold indicates the better window for each model.

	MSE ($\times 10^{-8}$)		MAE ($\times 10^{-4}$)		QLIKE	
Model	252	512	252	512	252	512
HAR	**30.12**	35.85	1.44	**1.27**	4.4254	**0.2573**
HAR-J	**30.16**	38.13	1.42	**1.26**	4.3471	**0.3303**
HAR-RS	**32.76**	34.74	1.53	**1.32**	7.1890	**0.6950**
HARQ	30.55	**28.85**	1.42	**1.28**	10.4502	**4.4127**
Log-HAR	27.19	**25.95**	1.26	**1.12**	0.3023	**0.1878**
ARFIMA	43.60	**39.75**	**1.31**	1.51	**0.2663**	0.3755